College, Debt, and Gender: The Influence of Outside Options on Higher Education

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Abstract

Women enroll in college at substantially higher rates than men, yet experience worse post-college outcomes, including lower earnings, lower labor supply, and higher student debt balances. This paper studies the role of gender differences in non-college labor market opportunities in shaping college enrollment, student debt, and aggregate gender inequality. We show that women's higher matriculation rates are concentrated in lower-income areas where non-college outside options are weaker, and that women disproportionately attend lower-quality, lower-cost institutions while still accumulating more student debt and earning less after graduation. To interpret these findings, we develop a model of college and career choice in which worse outside options raise the net return to college, increasing enrollment and debt even when post-college earnings remain lower. The framework highlights how changes in non-college labor markets can generate opposing movements in female welfare, the college earnings premium, and aggregate gender earnings gaps through compositional effects, with important implications for higher education and student debt policy.

Keywords: student debt, gender, outside options, higher education
JEL Codes: [TBD]

1 Introduction

U.S. student debt rose from $260 billion in 2004 to $1.5 trillion by 2019 (Federal Reserve Bank of New York, 2019), and forty-three percent of college attendees incur education-related debt (Federal Reserve Board of Governors, 2019). Two patterns stand out when debt and enrollment are broken down by gender. First, women are considerably more likely than men to attend and complete college (Goldin et al., 2006; Parker, 2021): in the cohorts we study, roughly forty percent of women hold a bachelor's degree by ages twenty-five to thirty, against roughly a third of men. Second, among those who attend, women borrow more: mean undergraduate borrowing among women exceeds men's by about $2,200 in 2018 dollars, a gap of roughly twenty-four percent (Addo and Kent, 2022; Chen and Smith, 2023). We refer to these two patterns together as the *twin gender gaps in higher education*. They are accompanied by worse financial outcomes after graduation, since women also earn less and supply fewer hours (Sloane et al., 2021).

The two gaps appear to point in opposite directions. Higher female attainment is among the largest and most durable improvements in women's relative position in the postwar economy, yet the women who obtained those degrees do not appear to have converted them into commensurate outcomes. This paper asks whether men's better non-college opportunities can account for both at once, and argues that much of the answer lies not in the college labor market but in the one women would face without a degree. The premise is that college is chosen against an outside option, so what a person forgoes matters as much as what she gains. A weaker outside option raises the net return to attending and draws more people into enrollment and into debt, even where the degree itself pays them comparatively little. If women's non-college prospects are worse than men's, then excess female matriculation, excess female borrowing, and disappointing post-college outcomes are not three puzzles but one equilibrium.

The paper makes three contributions. First, it documents the empirical configuration the twin gaps present, and shows that excess female matriculation is concentrated where the outside option is weakest and the ability to pay is lowest. Second, it provides a causal estimate of how sharply matriculation responds to the outside option, recovered by placing three existing quasi-experimental literatures on a common footing. Third, it develops a quantitative lifecycle model disciplined by that estimate, and uses it to show that two aggregates often read as measures of gender progress, the earnings gap and the college earnings premium, are composition-laden enough to reverse sign without any change in relative pay.

Empirical evidence. Section 2 establishes the labor market asymmetry at the center of the mechanism. Using the Current Population Survey, we estimate gender earnings

gaps separately in the college and non-college markets. The non-college gap is large and persistent, roughly twenty-four log points after controlling for year, race, age, geography, occupation, and industry, and it is not confined to male-dominated occupations: it is negative within twenty-one of twenty-six broad occupation groups, including those in which non-college women are concentrated. Measured on a clean split of exactly a high-school diploma against exactly a bachelor's degree, and in hourly terms, the implied proportional wedge is larger in the non-college market, $\tau_h = 0.172$, than in the college market, $\tau_c = 0.132$.

The consequences follow the geography of that floor. Excess female matriculation is disproportionately supplied by places and families with the weakest outside options and the least ability to pay. Across counties, women's enrollment exceeds men's by about twenty-eight percent at the bottom of the income distribution against about twenty-four percent at the top, a modest and non-monotone gradient, and the ratio is highest in the counties where the local non-college gender earnings gap is largest. Within families the proportional pattern is sharper: in the National Longitudinal Survey of Youth 1997, women in the bottom quintile of parental net worth are more than twice as likely as men to hold a degree by age thirty, against a ratio of 1.25 in the top quintile. These students then sort into the lower reaches of the school quality distribution, where women are sixty-one to sixty-three percent of enrollment against near parity at the most selective institutions, and where graduates earn roughly \$48,000 ten years after entry against \$91,000 at the top. The marginal female college student is therefore drawn disproportionately from households for whom attending requires borrowing, and enrolls where degrees pay least.

Causal evidence. The descriptive evidence is consistent with the mechanism but does not identify it, since the earnings available to non-college workers covary with much else that moves enrollment. To discipline the mechanism, we recreate three literatures that have isolated plausibly exogenous shocks to the local non-college labor market, the mid-2000s housing boom (Charles et al., 2018), the post-2007 fracking boom (Cascio and Narayan, 2022), and the rise in Chinese import competition (Autor et al., 2013; Ferrière et al., 2022; Conlisk et al., 2022). We place all three on a common outcome and a common endogenous regressor, the log of non-college expected income, so that their estimates can be read on one scale. The preferred estimate, from shale-reserve exposure at the commuting-zone level, is -0.190: a one percent rise in the non-college outside option lowers the enrollment rate of eighteen to twenty-four year olds by about 0.19 percentage points, an elasticity of roughly -0.50. The three headline designs span -0.129 to -0.190 despite differing in shock, geography, and era, and the China shock, the only one that moves the outside option downward, lands beside the two booms, so the enrollment response is not obviously asymmetric in the sign of the shock. Because the shocks bundle other channels and their compliers differ, we

read the three as a range rather than as replications of a single parameter.

Model and results. Sections 4 and 5 develop and calibrate a stationary lifecycle model of college and career choice in which the value of not attending is a central primitive. Households differ in ability and parental wealth, choose whether to attend and at what quality, borrow subject to a statutory cap and a means-tested grant schedule, may repay on an income-driven schedule following Ji (2021), and invest in human capital on the job through a Ben-Porath (1967) technology. Wages for college and high-school labor clear a CES labor market, which is held fixed in the calibration and allowed to adjust in the counterfactuals. The taste-shock scale is disciplined directly by the fracking estimate rather than left to absorb aggregate enrollment, which is a deliberate feature of the calibration: the central behavioral margin is pinned to a quasi-experimental moment and the rest of the model accommodates it.

We use the quantitative model to do counterfactuals. Removing the outside-option wedge alone pushes the matriculation gap below one, to 0.98, confirming that the marginal female attender is one for whom college is largely an escape from a penalized non-college market. Removing both wedges, so that women are paid exactly as identical men in both sectors, raises the aggregate female-to-male earnings ratio to 1.24. Under gender-symmetric pay, in other words, the model implies that women would out-earn men in aggregate by nearly a quarter, purely because more of them hold degrees. The aggregate gender earnings gap can therefore reverse sign without any woman being paid more than an identical man, and comparable forces move the college earnings premium, whose measured response in these experiments is dominated by the equilibrium price adjustment rather than by any change in what college teaches. Neither statistic should be read as a measure of relative wellbeing.

The remainder of the paper proceeds as follows. Section 2 presents the empirical motivation and the causal evidence. Section 3 develops a stripped-down model that delivers the central comparative statics in closed form. Sections 4 and 5 present and calibrate the quantitative model, and Section 6 reports the counterfactual decomposition. Section 7 concludes.

2 Empirical Motivation

Section 2.1 describes the data. We then document the labor-market asymmetry at the center of the mechanism, a gender earnings gap that is wider in the non-college market than in the college market, so that the return to a degree is higher for women. We trace its consequences: excess female matriculation is concentrated among lower-income families and localities, where the non-college outside option is weakest, and these students sort disproportionately into

lower-quality institutions, where the gender gap in student debt is largest. Finally, we provide causal evidence, from three research designs exploiting different shocks, that a weaker non-college outside option raises college enrollment.

2.1 Data

Table 1 lists the datasets, the years used, and what each supports. Sample restrictions, variable definitions, and geographic crosswalks are in Appendix B.

Table 1: Data sources

Source	Years	Used for
Descriptive evidence		
CPS ASEC	2004–2022	Gender earnings gaps by education
College Scorecard	2002–2023	School-level debt gaps, cost, earnings
IPEDS	2002–2023	Institutional characteristics, enrollment
NLSY97	1997–2023	Attainment by parental wealth
Causal designs		
Census and ACS	1990–2024	College enrollment
QWI	2000–2019	Non-college expected income
EIA	2008–2015	Shale reserves, play boundaries
SEER	2000	Population denominators
CHN, ADH packages	1990–2007	Housing and China-shock designs

Notes. CPS ASEC and Census/ACS are IPUMS extracts (Flood et al., 2024; Ruggles et al., 2024). Other sources are the College Scorecard (U.S. Department of Education, 2024), IPEDS, the NLSY97 (Bureau of Labor Statistics, U.S. Department of Labor, 2024), the Quarterly Workforce Indicators (U.S. Census Bureau, 2026), the Energy Information Administration (U.S. Energy Information Administration, 2015), SEER population estimates (Surveillance, Epidemiology, and End Results (SEER) Program, 2024), and the replication packages of Charles et al. (2018) and Autor et al. (2013). Census and ACS years are 1990, 2000, and 2005 through 2024. The Scorecard reports gendered debt medians only through 2021, which ends the debt-gap panel; other Scorecard variables run to 2023. NLSY97 respondents are the 1980–84 birth cohort, with parental net worth measured in 1997 and degree completion observed in rounds 17 through 21 (survey years 2015 through 2023). Sample restrictions and variable definitions are in Appendix B.

One construction recurs throughout and is worth stating here, since the causal estimates of Section 2.5 are reported as elasticities with respect to it. The outside option is measured as *expected income*, the average earnings of workers with a high-school education or less scaled by the employment rate of that same group. It is a price times a quantity, capturing both what non-college work pays and how likely it is to be found, rather than wages alone.

2.2 Gender Earnings Gaps in College and Non-College Labor Markets

We document gender earnings gaps in college and non-college labor markets using data from the Current Population Survey Annual Social and Economic Supplement (CPS ASEC). The sample pools survey years 2004 through 2022 and is restricted to full-time prime-age workers aged 25–54 with at least a high school degree. Annual earnings are deflated to 2022 dollars using the CPI-U. Sample summary statistics are reported in appendix Table 10.

To decompose earnings differences across gender and education groups, we estimate the regression

$$\log w_{it} = \beta_0 + \beta_1 \,\text{college}_i + \beta_2 \,\text{female}_i + \beta_3 \,(\text{female}_i \times \text{college}_i) + \mathbf{X}_{it}'\boldsymbol{\delta} + \varepsilon_{it} \tag{1}$$

where $\mathbf{X}_{it}$ includes progressively richer sets of fixed effects for year, race, age, geography, occupation, and industry. Under this specification, $\hat{\beta}_2$ identifies the non-college gender earnings gap relative to male non-college workers, $\hat{\beta}_1$ is the male college wage premium, and $\hat{\beta}_1 + \hat{\beta}_3$ is the female college wage premium. Table 2 reports estimates across five specifications. Three findings stand out.

First, there is a large non-college gender earnings gap. In the raw specification (column 1), non-college women earn approximately 31 log points less than non-college men ($\hat{\beta}_2 = -0.311$). This gap narrows to 24 log points after all controls are included ($\hat{\beta}_2 = -0.245$, column 5) but remains large and highly significant throughout. Second, in the raw specification the college earnings premium is nearly identical for men ($\hat{\beta}_1 = 0.557$) and women ($\hat{\beta}_1 + \hat{\beta}_3 = 0.548$). Third, $\hat{\beta}_3$ is positive throughout and grows as occupation and industry fixed effects are added, reaching 0.045 in the full specification. The interaction term thus reveals a differential college return for women that is not apparent in the raw data and that operates through occupational sorting: conditioning on where workers are employed shows that college delivers a larger wage premium for women than for men.

The remainder of this subsection turns from the regression coefficients to the objects the model's gender wedges correspond to: the implied gender earnings wedges $\tau_h = 1 - \exp(\hat{\beta}_2)$ in the non-college market and $\tau_c = 1 - \exp(\hat{\beta}_2 + \hat{\beta}_3)$ in the college market. These are estimated on a deliberately clean education split, workers holding exactly a high-school diploma against exactly a bachelor's degree ($N = 521{,}416$), rather than the broad associate-based split of Table 2. The clean split is used because it keeps the wedges from being diluted by associate degrees and incomplete college. It is not identical to the education split used to construct the model's calibration targets, which take college to be a bachelor's degree or higher; Section 5

Table 2: Gender Wage Gaps by Education

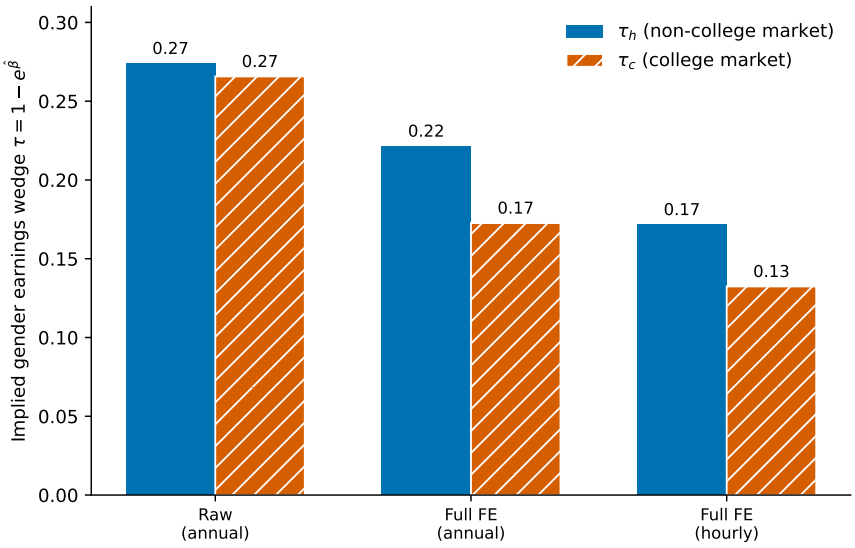
	(1)	(2)	(3)	(4)	(5)
Female	-0.311***	-0.310***	-0.259***	-0.295***	-0.245***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
College	0.557***	0.507***	0.236***	0.417***	0.217***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Female × College	-0.009**	0.016***	0.045***	0.051***	0.045***
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Year FE	Yes	Yes	Yes	Yes	Yes
Demo. and geo. FE	No	Yes	Yes	Yes	Yes
Occupation FE	No	No	Yes	No	Yes
Industry FE	No	No	No	Yes	Yes
Observations	943,171	943,171	943,171	943,171	943,171
R-squared	0.146	0.201	0.334	0.266	0.353

Notes: Dependent variable is log real annual wages (2022 dollars). Each column adds controls incrementally: (1) year FE; (2) adds race, age, state, and county FE; (3) adds occupation FE; (4) adds industry FE; (5) full specification with occupation and industry FE. Weighted by ASEC person weights; standard errors are robust to heteroskedasticity. Asterisks denote statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Sample: full-time workers aged 25–54, CPS ASEC 2004–2022.

discusses the measurement choices behind those targets. The choice matters for what follows, so we flag it here rather than in a note.

Figure 1 plots the resulting wedges. In the raw annual data the two are nearly equal (0.274 against 0.266): unconditionally, women face a similar proportional earnings shortfall in both markets. Moving to the full set of fixed effects, and then to hourly wages, both wedges shrink and the ordering that matters for the model emerges clearly: τ_h exceeds τ_c by 4.9 percentage points with full controls (0.221 against 0.173) and by 3.9 points in hourly terms (0.172 against 0.132), the labor-supply-purged values closest to the model's price wedges. The reason the raw wedges conceal this ordering, a college gender gap concentrated in the top tail of earnings, is taken up below.

Figure 1: Gender Earnings Wedges in College and Non-College Markets

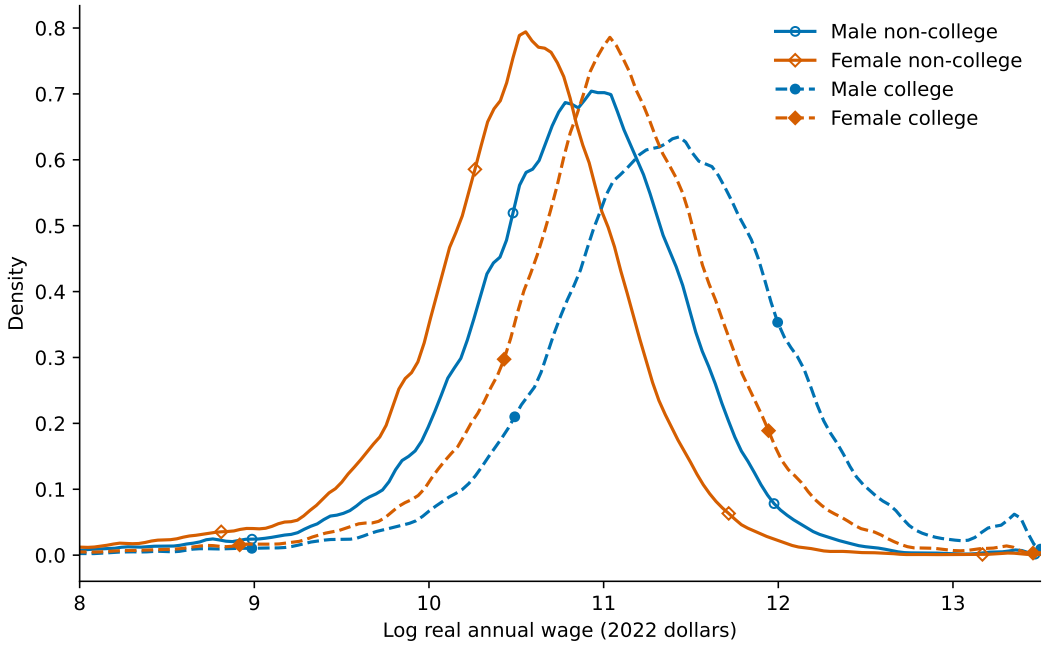


Notes: Implied wedges $\tau = 1 - \exp(\hat{\beta})$ from equation (1): τ_h from $\hat{\beta}_2$, τ_c from $\hat{\beta}_2 + \hat{\beta}_3$. Raw: year fixed effects only. Full FE: year, race, age, state, county, occupation, and industry fixed effects. Hourly divides annual earnings by weeks worked and usual weekly hours. Weighted by ASEC person weights. Confidence intervals are smaller than the printed value labels at these sample sizes; standard errors are reported in Table 3. Sample: full-time workers aged 25–54 holding exactly a high-school diploma or exactly a bachelor's degree, CPS ASEC 2004–2022 ($N = 521{,}416$).

Figure 2 plots survey-weighted kernel densities of log annual wages for each of the four gender-by-education groups. The non-college distributions are shifted apart throughout their body: the male distribution lies well to the right of the female across the central mass. The college distributions track each other more closely through the middle of the distribution, and their separation is concentrated in the upper tail, where the male college density carries

substantially more mass, the same top-tail asymmetry documented below in the discussion of Table 3. The contrast, a broad wage-floor disadvantage for non-college women against a top-concentrated gap for college women, is consistent with the regression evidence in Table 2 and motivates the outside-option argument developed in Section 3.

Figure 2: Wage Distributions by Gender and Education



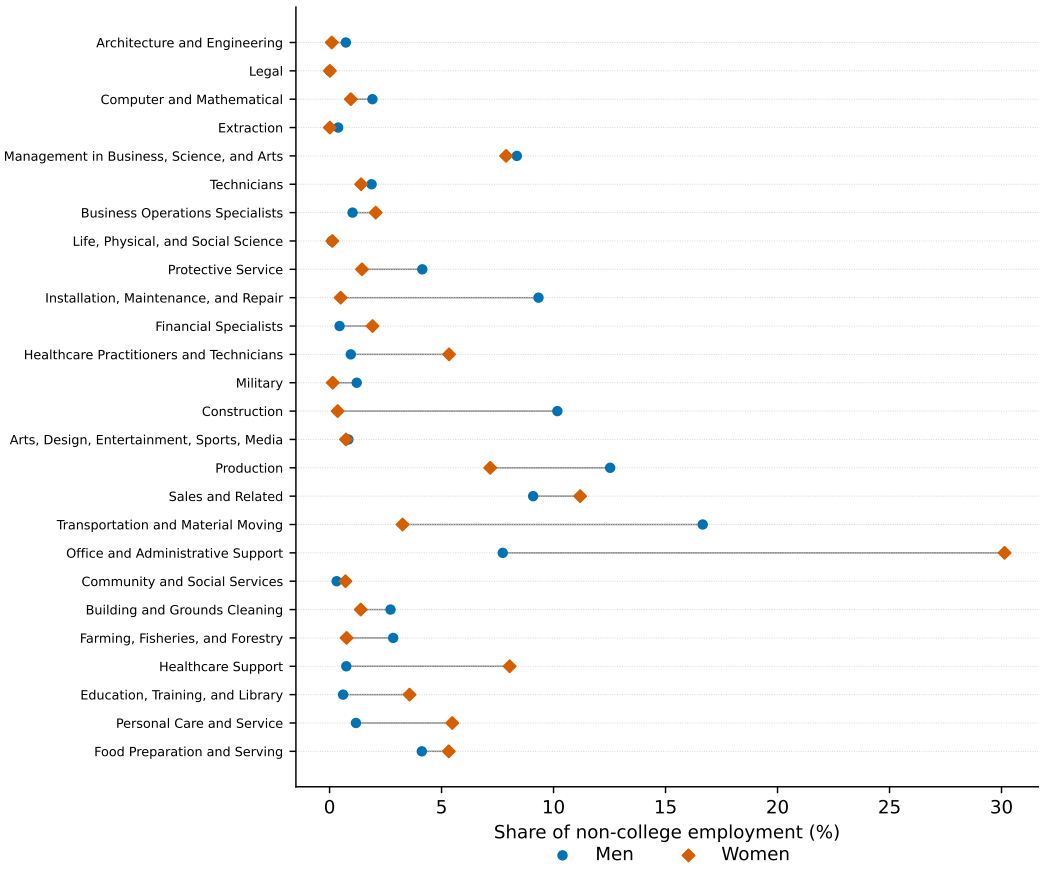
Notes: Survey-weighted kernel densities of log real annual wages for four gender-by-education groups. Sample: full-time workers aged 25–54, CPS ASEC 2004–2022. Non-college denotes less than an academic associate degree (IPUMS educ below 92; vocational associate degrees fall in the non-college group).

The divergence is visible directly in Appendix Figure 13, which plots the two premiums across specifications on the 2004–2022 sample: nearly identical in the raw specification, with the female premium exceeding the male by about 4.5 log points (0.262 against 0.217) once occupation and industry are controlled. The same divergence appears in the earlier 1995–2003 window.

To understand the mechanism behind the larger non-college gap, Figures 3 and 4 examine its occupational dimension. Figure 3 plots the share of non-college employment in each broad occupation group for men and women separately, with occupations sorted from bottom to top by average non-college log wage. Setting aside the professional groups at the very top, which apart from Management employ few non-college workers of either sex, non-college employment for men is concentrated in the well-paying blue-collar groups in the

upper middle of the chart: Transportation and Material Moving, Production, Construction, and Installation, Maintenance, and Repair, the last two of which employ almost no women. Non-college employment for women is concentrated in Office and Administrative Support and in the service groups at the bottom: Healthcare Support, Personal Care and Service, and Food Preparation and Serving. Non-college men therefore have access to a set of high-wage occupations that are largely unavailable to non-college women.

Figure 3: Occupational Sorting Among Non-College Workers



Notes: Share of non-college employment in each occupation group for men (blue circles) and women (vermillion diamonds). Occupations sorted from bottom to top by average non-college log wage. Occupation groups approximate the 2010 SOC categories using occ1990 codes (the extract lacks occ10ly; the exact mapping returns with the new extract). Survey-weighted. Sample: non-college full-time workers aged 25–54, CPS ASEC 2004–2022.

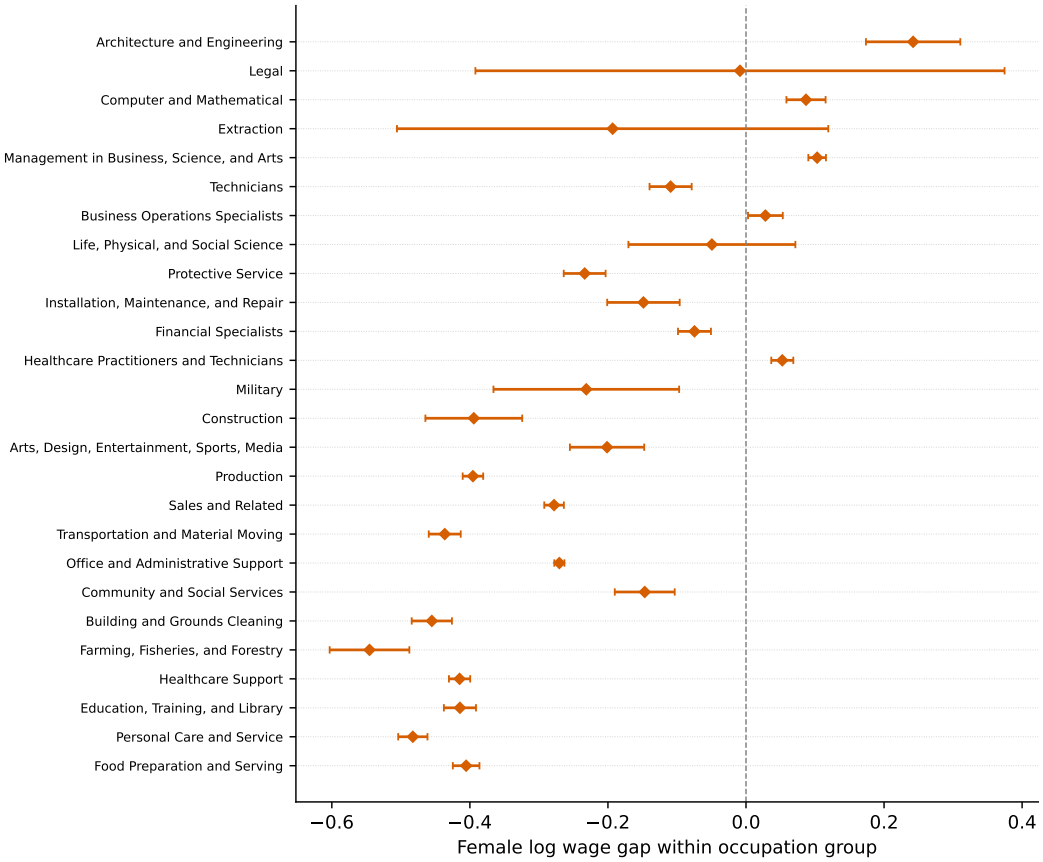
Figure 4 asks whether the non-college gender gap is confined to male-dominated occupations or is more broadly present. It plots the estimated female wage gap within each of the 26 broad occupation groups, restricting the sample to non-college workers and controlling for

year, geography, race, age, and industry fixed effects but not occupation fixed effects, so that the within-group gap is recovered directly for each occupation. The gap is negative in 21 of the 26 groups, spanning every blue-collar and service group in which non-college employment is concentrated, with gaps of up to roughly 55 log points. Even in female-dominated groups, including Healthcare Support and Personal Care and Service, non-college women earn significantly less than non-college men in the same broad occupation. The five exceptions are the professional and managerial groups, where non-college women earn more than non-college men within group (by 3 to 24 log points); these groups are not uniformly small (Management alone accounts for roughly 8 percent of non-college employment), and the reversal plausibly reflects strong selection of non-college workers into professional occupations. We flag these five groups as provisional: their sign is not stable across the occupation coding and sample window, and the pattern in the blue-collar and service groups, where non-college employment is actually concentrated, is what the argument rests on. We do not interpret this as evidence that occupational sorting is irrelevant: Figure 3 makes clear that sorting into high-wage occupations is an important part of the story. Rather, the within-occupation gaps indicate that non-college women face worse earnings prospects than non-college men across the full occupation distribution, and not only on the margin of access to male-dominated industries.

Two further results connect these earnings gaps to the wedges the quantitative model consumes. First, part of the measured gap operates through labor supply rather than wage rates. Table 3 re-estimates the gaps under three earnings concepts: annual earnings, weekly wages, and hourly wages. The non-college wedge falls from 0.274 (annual, raw) to 0.223 (hourly, raw), and the fully controlled hourly wedges are $\tau_h = 0.172$ and $\tau_c = 0.132$. Under full controls τ_h exceeds τ_c by 3.9 to 4.9 percentage points in every concept. The hourly wedges are the model-relevant objects, since the model treats hours as a choice and the wedge as a price.

Second, the ordering $\tau_h > \tau_c$, which the model interprets as college insuring women against a weak non-college market, is a statement about the body of the earnings distribution rather than its top. In the raw data the two wedges differ by only 0.8 to 1.3 percentage points depending on the concept, and median-based wedges are equal to three decimal places (0.260 against 0.260), because the college gender gap is concentrated among top earners: excluding the top one percent of the within-year earnings distribution widens the raw annual gap between τ_h and τ_c from +0.008 to +0.028, and excluding the top five percent widens it to +0.074, with τ_c falling sharply and τ_h nearly unchanged. On the broad associate-based split of Table 2 the raw ordering reverses slightly, which is itself the top-tail point in another guise. The college market thus exhibits a glass-ceiling pattern, a gender gap concentrated at the very top, while the non-college gap is spread throughout the distribution, precisely

Figure 4: Non-College Gender Wage Gap by Occupation Group



Notes: Estimated female wage gap within each broad occupation group, non-college workers only. Estimated from a regression of log real wages on occupation-group-by-female interactions, controlling for year, geography, race, age, and industry fixed effects. No occupation fixed effect is included, so the gap is recovered separately within each group (occ1990 codes grouped to approximate the 2010 SOC categories). Brackets are 95% confidence intervals. Sample: non-college full-time workers aged 25–54, CPS ASEC 2004–2022.

the wage-floor disadvantage that makes the outside option weak for the typical non-college woman. Table 3 reports the full ladder across earnings concepts, and its note gives the broad-split values.

The ladder extends naturally to the labor supply margins that the full-time sample removes, completing the decomposition. Staying on the clean high-school-versus-bachelor's split and the raw specification, the non-college wedge is 0.22 in hourly wages among full-time workers, 0.27 in annual earnings among full-time workers, 0.34 in annual earnings among all labor force participants with positive earnings (adding the part-time and part-year margin), and 0.44 in mean annual earnings including zeros for the entire population with no work

Table 3: Implied Gender Earnings Wedges by Earnings Concept

	Annual		Weekly		Hourly	
	Raw	Full FE	Raw	Full FE	Raw	Full FE
τ_h (non-college wedge)	0.274	0.221	0.262	0.199	0.223	0.172
τ_c (college wedge)	0.266	0.173	0.249	0.158	0.212	0.132
Observations	521,416					

Clean education split: exactly a high-school diploma versus exactly a bachelor's degree. On the broad (academic-associate-or-higher) split of Table 2 (N = 943,171), the corresponding values are annual 0.267/0.274 raw, 0.217/0.181 full FE; weekly 0.253/0.257 raw, 0.194/0.165 full FE; hourly 0.212/0.218 raw, 0.165/0.138 full FE (τ_h/τ_c).

Notes: Implied wedges $\tau = 1 - \exp(\hat{\beta})$ from equation (1): τ_h from $\hat{\beta}_2$, τ_c from $\hat{\beta}_2 + \hat{\beta}_3$. Raw: year fixed effects only. Full FE: year, race, age, state, county, occupation, and industry fixed effects. Weekly divides annual earnings by weeks worked; hourly additionally divides by usual weekly hours. Weighted by ASEC person weights. Sample: full-time workers aged 25–54 holding exactly a high-school diploma or exactly a bachelor's degree, CPS ASEC 2004–2022; the table's own note reports the broad-split values.

conditioning at all (adding the participation margin); the college wedge climbs in parallel from 0.21 to 0.42.[1] Each step isolates one margin: hours, weeks and part-time work, and participation. The participation margin is sizable in both markets, and is what carries the final step: the share of women with positive annual earnings is 11.7 percentage points below men's among high-school workers (0.671 against 0.788) and 10.2 points below among bachelor's holders (0.792 against 0.894). Two implications follow. First, the fully inclusive gender earnings gap is nearly education-neutral, roughly 42 to 44 percent in both markets: the ordering $\tau_h > \tau_c$ is a statement about wage rates conditional on working, below the top tail, not about total earnings. Second, the mapping to the model is now explicit at both ends of the ladder: the model consumes the hourly price wedges at the top, and it must endogenously reproduce something close to the inclusive 0.42-to-0.44 gap at the bottom through its hours and participation choices, which makes the bottom rung an untargeted validation moment rather than an input.

Taken together, the evidence in this section documents three facts. First, there is a large and persistent non-college gender earnings gap of approximately 24 log points after full controls, reflecting both differential sorting into occupations and within-occupation disadvantage for non-college women. Second, the college earnings premium is higher for women than for

[1] The final step cannot be estimated in logs; it is a Poisson pseudo-maximum-likelihood regression of real annual earnings levels, including zeros, on the same regressors, whose coefficients retain the proportional interpretation.

men once occupation and industry are controlled, with women gaining approximately 4 to 7 log points more from a college degree. Third, both patterns appear with somewhat larger magnitudes in the earlier 1995–2003 window ($\hat{\beta}_2 = -0.380$ raw and -0.321 with full controls; $\hat{\beta}_3 = 0.067$) as well as in the 2004–2022 sample, so they are stable across time and are not an artifact of the sample period. These facts motivate the model in Section 3, in which college serves not only as a means to acquire skills but also as a way to access labor market opportunities that are comparatively poor for women without a college degree.

2.3 Sources of Excess Female Matriculation

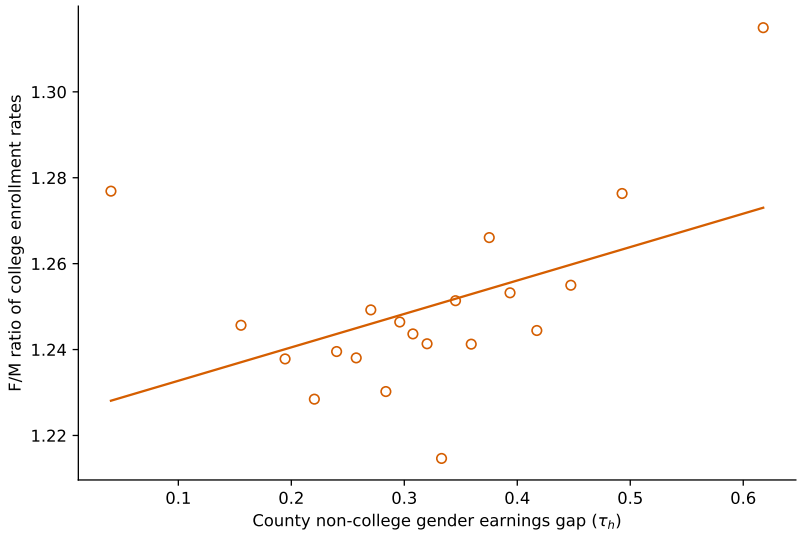
Given the aggregate twin gender gap in higher education, a natural question is where this excess female matriculation (relative to men) comes from. Does the population of female college students come from the same localities and families as male college students only in great numbers, or is there a systematic difference in male and female matriculation also at the local, micro level? The answer can shed light on the economic forces that drive the differences in male and female matriculation decisions and their economic implications.

Figures 5 and 6 provide insight by showing how the gender gap in college enrollment varies across U.S. counties with different economic characteristics. The figures are constructed from the American Community Survey (2005–2024, together with the 2000 Census long form). Enrollment is measured as the share of residents aged 18 to 24 currently attending college, computed separately by sex within each county and year, and the outcome is the county female-to-male ratio of these enrollment rates. The local non-college gender earnings gap (τ_h) is the log difference between male and female mean wage income among high-school-educated workers aged 25 to 55 in the county, and the county income percentile ranks counties within each year by mean worker income, weighted by population. Both binscatters residualize year fixed effects and county demographic controls (racial composition, education shares, age structure), weight by county population, and exclude county-sex cells below a minimum population threshold; the threshold is not consequential, dropping four of 8,642 county-years and moving the slope by less than a tenth of a percent.

Figure 5 shows that excess female matriculation comes disproportionately from localities in which women face worse outside (e.g., non-college) options. It plots the county female-to-male enrollment ratio against the local non-college gender earnings gap. The ratio is roughly 1.28 in the counties with the smallest non-college gender earnings gaps, dips modestly through the middle of the distribution (reaching 1.21 near the median), and rises to roughly 1.31 in the counties with the largest gaps, so the positive relationship is concentrated in the upper part of the τ_h distribution. The fitted slope is 0.078 (standard error 0.031, clustered

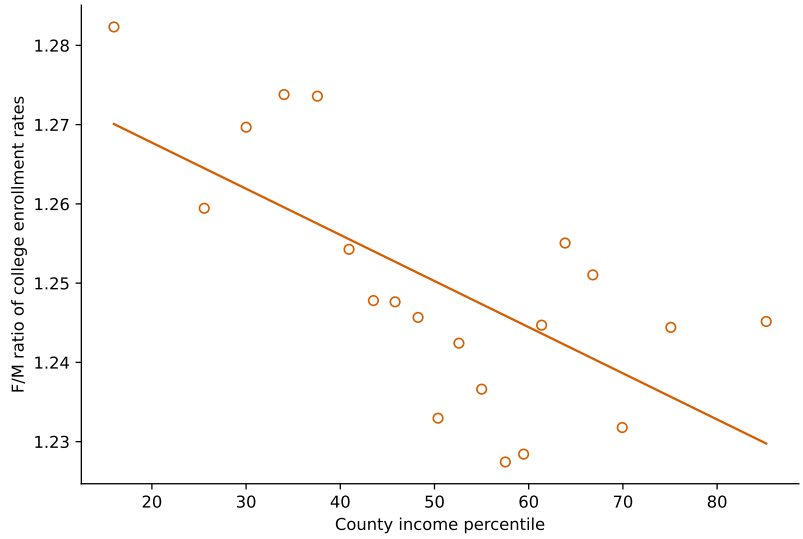
by county): a one-log-point wider local non-college gender earnings gap is associated with an enrollment ratio higher by about 0.08, roughly the distance between the middle and the top of the binned series. The result suggests that differences in local non-college labor market opportunities may be part of the reason that women seek college education at greater rates than men.

Figure 5: Enrollment Ratios and Local Non-College Labor Market Earnings Gaps



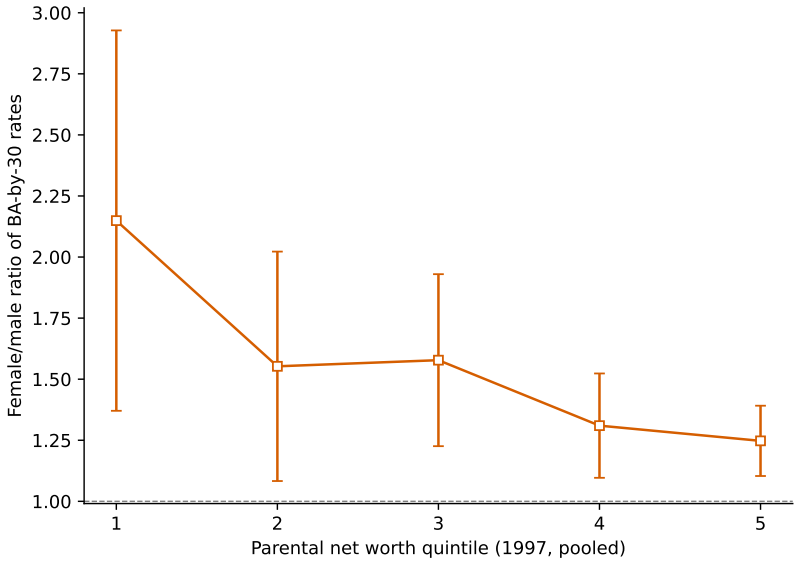
Notes: American Community Survey 2005–2024 and 2000 Census, county-year panel (515 counties with identifiable county codes and sufficient cell sizes). Twenty population-weighted bins; year fixed effects and county controls residualized.

Figure 6: Enrollment Ratios and County Income Percentiles



Notes: American Community Survey 2005–2024 and 2000 Census, county-year panel. Twenty population-weighted bins; year fixed effects and county controls residualized.

Figure 7: Attainment Ratio by Parental Wealth



Notes: National Longitudinal Survey of Youth 1997 (cohort born 1980–84), custom longitudinal weights, final sample 5,913. The figure plots the female-to-male ratio of bachelor's completion rates by age 30 across pooled quintiles of parental household net worth measured in 1997, before the college decision. Brackets are 95 percent delta-method confidence intervals.

Figure 6 displays another dimension of county characteristics that is highly correlated with excess female matriculation. It plots the county female-to-male enrollment ratio against county income percentiles. The correlation is negative but modest, and the binned series is not monotone: women's enrollment exceeds men's by roughly 28 percent in the poorest counties, reaches its minimum of roughly 23 percent just above the middle of the distribution, and ends at roughly 24 percent at the top. The fitted slope is -0.0006 per percentile (standard error 0.0003, clustered by county), marginally significant. The result can help explain why the student debt gap between men and women has grown with the matriculation gap: excess female matriculation comes disproportionately from low-income localities where paying for college is more likely to require taking on debt.

Figure 7 reinforces this observation by showing a similar relationship within families ranked by wealth. Using the National Longitudinal Survey of Youth 1997, it plots the female-to-male ratio of bachelor's completion by age 30 across quintiles of parental net worth measured in 1997, before the college decision. Since the NLSY 1997 is a household survey, the sample explicitly includes siblings in the same families which, by construction, fall into the same parental wealth bins. As at the county level, the gender gap is proportionally larger in lower-wealth households: in the bottom wealth quintile, women are more than twice as likely as men to hold a degree by age 30 (a ratio of 2.15), and the ratio declines to 1.25 in the

top quintile. Since poor and wealthy households are equally likely to have children of either sex, the result suggests that wealthy households send their sons and daughters to college at roughly similar rates, while in lower-wealth households daughters are roughly twice as likely to earn a degree as sons. The two readings of this exhibit differ, and we report both. In percentage points the female excess is roughly flat to mildly increasing across quintiles, between 6.7 and 12.7 points, so the concentration in low-wealth households is a statement about proportions rather than levels. We emphasize the ratio because a given percentage-point gap represents a far larger behavioral difference where baseline attendance is low, but the level reading is the one that speaks to how many additional female borrowers the bottom quintiles supply.

2.4 Destination of Excess Female Matriculation

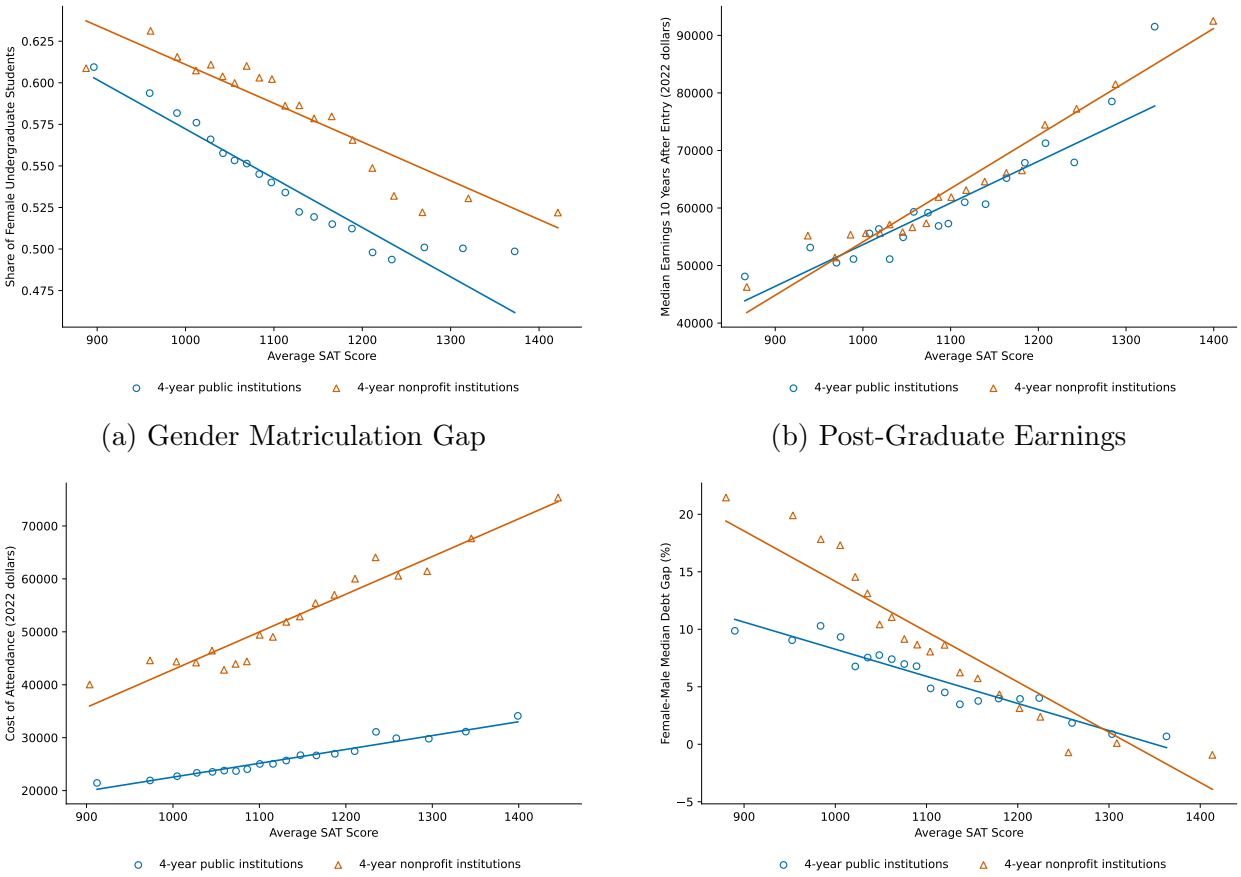
Having established where the additional female students come from, we turn to where they end up. Is excess female matriculation spread evenly across institutions, or concentrated in particular types of school? The answer bears directly on the consequences of the twin gender gap, since where a student enrolls determines both what she pays and what she earns afterward.

The four panels of Figure 8 summarize differences in the sorting of men and women across colleges. The primary data comes from the National Center of Education Statistics (NCES) IPEDS, an administrative database covering the universe of accredited higher education institutions in the United States.

The first sub-figure 8a shows that the gender imbalance among college enrollees is far from uniform across schools. It plots the share of undergraduate students who are female across the college quality distribution, proxied by the average SAT scores of the study body, for both public and private non-profit institutions. The result shows that excess female matriculation is almost entirely concentrated among lower quality schools. In the highest quality colleges, there is a nearly perfect gender balance across both public and private institutions with women representing 50% of the student body, or just over that at private non-profit schools. In contrast, female students account for roughly 61 to 63% of the student body at the lowest quality colleges.

These gender disparities in college sorting have implications for the expected returns to college and post-graduate earnings of men and women. This is because, as Figure 8b shows, graduates from lower quality colleges have lower expected earnings. On average, graduates from the highest quality colleges earn nearly twice as much as those from the lowest quality colleges 10 years after entry (e.g., \$91k versus \$48k). Hence, the more excess female

Figure 8: Student Sorting in Higher Education, by Public-Private Sector



(a) Gender Matriculation Gap

(b) Post-Graduate Earnings

(c) Cost of Attendance

(d) Student Debt Gap

Notes: Source data comes from the National Center of Education Status (NCES) Integrated Postsecondary Data System (IPEDS) and the College Scorecard.

matriculation becomes concentrated at lower quality schools, the larger gender disparities become in the returns to schooling. This in turn has implications for aggregate measures of income inequality, such as the gender earnings gap and the college earnings premium.

An important question is whether these gender differences in sorting across colleges is the result of discrimination by colleges or other unobserved differences between male and female applicants. For example, colleges may have an inherent preference for gender balance among their study body despite larger numbers of female applicants. Alternatively, with strong selection into college, the marginal female applicants may be lower ability or less prepared for admission to top colleges. While the micro data do not allow us to definitely differentiate between the explanations, aggregated data suggest the differences in sorting are

not the result of differences in aptitude. For example, according to the National Center for Education Statistics (2024), male test takers outscored women on the SAT by an average of only 9 points (1032 versus 1023) in 2023. Given the small national gender gap in test scores relative to the large variation across schools, it is unlikely that differences in aptitude explain the gender differences in college sorting.[2]

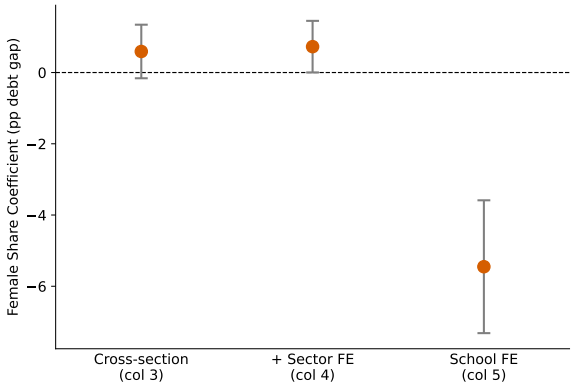
The data do suggest that the ability to pay for higher-quality colleges plays a role. As the preceding subsection showed, the marginal female matriculators are more likely to come from low-income families, so they may enroll in lower-quality colleges partly because those colleges cost less. As Figure 8c shows, the cost of attendance rises with college quality in both sectors, nearly doubling between the lowest- and highest-quality nonprofits and rising by about sixty percent across publics. Yet despite disproportionately attending these cheaper schools, female students still graduate with more debt than their male peers, as Figure 8d illustrates. At least part of the sorting difference therefore appears to be driven by the ability to pay.

However, the data suggest that the differences in sorting are not merely the result of ability to pay in a static sense. Regression analysis of the school-level debt gap reveals that the cross-sectional relationship between female enrollment share and the debt gap largely reflects differences in school characteristics, particularly school quality, rather than female composition per se. As Figure 9 shows, controlling for school fixed effects, years in which a school enrolls a larger female share are associated with *smaller* debt gaps, not larger ones, the opposite of the cross-sectional pattern. This pattern is consistent with universities actively responding to gender imbalance in their applicant pools by offering more generous aid packages to male students, a mechanism that would compress the gap as female enrollment rises.

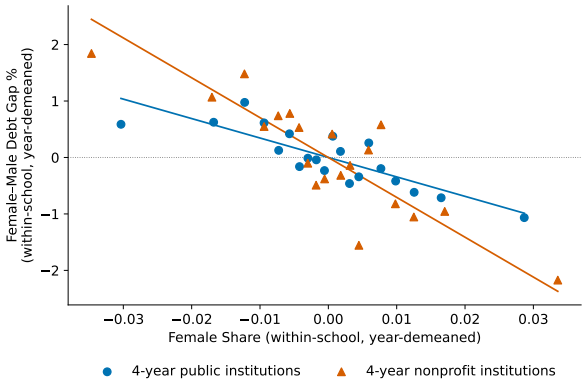
Taken together, the evidence suggests that the female debt gap in Figure 8d is primarily a structural feature of lower-quality institutions, where women disproportionately enroll, rather than a direct consequence of female enrollment itself. The underlying sources of this gap likely include the lower parental assets of the marginal female college attendee documented in the preceding subsection, as well as institutional financial aid behavior that responds to the gender composition of applicants. The model of Section 4 speaks to the first of these, since it carries parental wealth and need-based aid explicitly; it has no institutional aid-response channel, so the second remains a hypothesis this paper documents rather than tests. Two caveats discipline this reading. First, the outcome is the gap in median

[2]Subscore gaps are similarly small relative to the variation across schools: men average 515 on Math against women's 500, while women outscore men on Evidence-Based Reading and Writing, 523 against 517. Men's scores do exhibit larger variance, which could contribute to sorting differences if selection on SAT scores across colleges is sufficiently strong.

Figure 9: Female Enrollment Share and the Debt Gap: Cross-Section vs. Within-School



(a) Coefficient on Female Share Across Specifications



(b) Debt Gap vs. Female Share (Within-School)

Notes: Left panel plots the coefficient on female enrollment share (standardized to mean 0, SD 1) from three specifications: cross-sectional with controls (col. 3), adding aid and admissions controls and sector fixed effects (col. 4), and school fixed effects (col. 5) of Table 11. Bars are 95% confidence intervals; standard errors clustered at the school level. Right panel plots the female–male median debt gap against female enrollment share, both residualized on school and year fixed effects. Enrollment-weighted. Estimation sample: 4-year public and nonprofit institutions, year_end 2010–2021, dictated by joint regressor availability (see Table 11; the debt-gap data themselves cover 2002–2021).

cumulative federal debt among borrowers, so it excludes the participation margin and can reflect composition differences between female and male borrowers within a school. Second, the facts are insensitive to sample construction: excluding specialized and tribal institutions everywhere and guarding the debt-gap denominator changes no sign or ranking.

2.5 Causal Evidence on Outside Options and Matriculation

The mechanism at the center of the paper is that a weaker non-college outside option raises the net return to college and draws more people into enrollment and into debt. The descriptive evidence above is consistent with this mechanism but does not identify it, because the earnings available to non-college workers are correlated with many local characteristics that also move enrollment. To discipline the mechanism, and in particular to pin down how sharply matriculation responds to the outside option, the paper needs a causal estimate of the effect of non-college earnings prospects on college enrollment. This subsection provides a range of such estimates.

Three literatures have already isolated plausibly exogenous shocks to the local non-college labor market and traced their effect on schooling: the mid-2000s housing boom (Charles et al., 2018), the post-2007 fracking boom (Cascio and Narayan, 2022), and the rise in Chinese import competition (Autor et al., 2013), extended to schooling by Ferrière et al. (2022) and, gender-split, by Conlisk et al. (2022). The housing boom raised demand for young non-college workers where housing supply was inelastic and reduced college attendance; the fracking boom raised demand for less-educated labor where shale lay underground and raised high-school dropout rates; and Chinese import competition *lowered* the earnings and employment of non-college workers in exposed commuting zones and raised college enrollment. We recreate all three in a single consistent form, taking the outcome to be a measure of college matriculation and the endogenous regressor to be the log of the non-college *expected income* constructed in Section 2.1, so that the three estimates can be read on one scale. This requires a small departure from each original: we replace the Cascio-Narayan high-school outcome with college enrollment, re-scale the housing design so the instrumented variable is the outside option rather than the housing shock itself, and re-scale the China-shock reduced form to the same expected-income coefficient. The two booms are positive shocks to the outside option and the China shock is negative, so the three together also test whether the enrollment response is symmetric in the sign of the shock. The aim is a common yardstick for a single number, the semielasticity of matriculation to the non-college outside option.

2.5.1 The ideal experiment

The object of interest is the coefficient θ in a regression of college matriculation on the expected income available without a degree,

$$M_i = \alpha + \theta \, \ln \mathrm{ExpInc}_i + \Gamma' X_i + \varepsilon_i, \tag{2}$$

where M_i is a measure of college matriculation for unit i, $\ln \mathrm{ExpInc}_i$ is the log non-college expected income of Section 2.1, and X_i collects controls. The coefficient θ is the semielasticity of matriculation to the outside option, and the paper's mechanism predicts $\theta < 0$: a stronger outside option lowers the net return to college and reduces enrollment. Throughout, M denotes a common object, the rate of college participation, and the three designs differ in how they measure and identify it. The fracking design identifies θ from an annual panel and enters M in levels; the housing design identifies it from a cross-sectional long difference and enters M as a change ΔM; and the China-shock design identifies it from a stacked panel of long differences, entering the enrollment flow as a change ΔM.

Estimating (2) by ordinary least squares does not recover θ. Expected income without

college is not assigned at random across people or places; it is correlated with local human capital, industry composition, school quality, and family resources, all of which independently move matriculation, so the OLS estimate confounds the effect of the outside option with these omitted factors, with a bias of ambiguous sign. Identification requires an instrument for $\ln \text{ExpInc}$ that meets two requirements. Relevance requires that the instrument shift the local non-college outside option, which is testable in the first stage. Exogeneity requires that it be unrelated to the determinants of matriculation except through the outside option, which is not directly testable and must be argued from the nature of the shock. The three designs supply three such instruments, a shale-reserve exposure, a housing-demand shock, and a supply-driven measure of Chinese import competition, and recover θ from otherwise unrelated variation.

2.5.2 The fracking design

Following Cascio and Narayan (2022), the first design uses the post-2007 fracking boom. A commuting zone's shale reserves are fixed by nature, and the drilling technology that made those reserves worth extracting arrived suddenly and everywhere at once. Zones sitting on more shale therefore saw a sharp rise in the demand for non-college labor, at a common moment and for a reason unrelated to their own economies. This is the source of exogenous variation. Because the geology is fixed and the boom's timing is common, a zone's exposure should move its college enrollment only through the labor-market shock it delivers, not through anything else specific to the zone. We estimate, at the CZ-by-year level, the first stage

$$\ln \text{ExpInc}_{c,t} = \lambda \left(\text{Reserves}_c \times \text{Post}_t\right) + \gamma_c + \delta_{s(c),t} + u_{c,t}, \tag{3}$$

and the second stage

$$M_{c,t} = \theta \, \ln \text{ExpInc}_{c,t} + \gamma_c + \delta_{s(c),t} + \varepsilon_{c,t}, \tag{4}$$

Here $M_{c,t}$ is the share of 18-to-24-year-olds in commuting zone c and year t who are enrolled in college, and expected income is instrumented by the interaction $\text{Reserves}_c \times \text{Post}_t$. This interaction switches on a zone's standardized per-capita shale reserves in the boom years ($t \geq 2008$), so it picks out the extra labor demand the boom delivered to high-shale zones. The two sets of fixed effects strip out confounds. The zone effects γ_c absorb the fixed geology and anything else constant about a zone, and the state-by-year effects $\delta_{s(c),t}$ absorb any statewide shock to enrollment or the labor market. Identification therefore comes only from differences, within a state and over time, between zones with more and less shale; these two sets of fixed effects are the only controls. Observations are weighted by each zone's year-2000 population aged 18 to 24, and standard errors are clustered by commuting zone.

The sample is the 311 zones in the seventeen states with a major shale play, over a boom window of 2003 to 2015; the sample definition, and how it differs from that of Cascio and Narayan (2022), is in Appendix B. The reported specification drops 2005, for a data reason given below. Because the ACS provides no sub-state geography before 2005, the second stage is estimated on 2006 through 2015 ($N = 3{,}110$), while the first stage, which uses the QWI, spans the full window ($N = 3{,}732$).

2.5.3 The housing-boom design

Following Charles et al. (2018), the second design uses the mid-2000s housing boom. When house prices rise, so does local demand for young non-college workers, above all in construction and related trades, and the effect is largest in metropolitan areas where housing supply is too constrained to meet demand by building more. The boom therefore gave a temporary lift to the non-college outside option, and gave the biggest lift where housing was tightest. The problem is that a metro area's housing boom is not random: places that boomed may differ in other ways that also shape who goes to college. To isolate the housing-specific part, Charles et al. (2018) instrument the local housing shock with the size of a *structural break* in the area's house prices. They fit each metro area's quarterly log house-price series over 2000 to 2005 with a linear pre-break trend plus a single one-time break, choosing the break date that best fits the series; the instrument Z_m is the magnitude of that break, meaning how sharply the growth rate of prices jumped once it occurred. The identifying assumption is that fundamentals such as local productivity or income feed into prices smoothly, so a sudden break from trend reflects housing-specific, speculative forces, such as shifting credit conditions or a price bubble, rather than the slower-moving local factors that could independently affect college-going. A bigger break predicts a bigger rise in local housing demand over the boom, and hence a bigger lift to the non-college outside option, which is what Z_m instruments here.

We estimate the design in long differences from the 2000 Census to pooled 2005–2007 ACS data (the endpoint Charles et al. (2018) label 2006), with the housing-demand shock measured over the 2000–2006 boom, at the metropolitan-area level, with first stage

$$\Delta \ln \text{ExpInc}_m = \lambda \, Z_m + X_m' \pi + u_m, \tag{5}$$

and second stage

$$\Delta M_m = \theta \, \Delta \ln \text{ExpInc}_m + X_m' \beta + \varepsilon_m, \tag{6}$$

where ΔM_m is the change in the any-college share of 18-to-25-year-olds, $\Delta \ln \text{ExpInc}_m$ is the

change in log expected income instrumented by Z_m, and the control vector X_m holds the four baseline covariates of Charles et al. (2018): the share of workers with a college degree, the female share of the labor force, the foreign-born share, and log metro-area population in 2000. We use the authors' population weights and cluster by state.

2.5.4 The China-shock design

Following the China-shock literature (Autor et al., 2013; Conlisk et al., 2022; Ferrière et al., 2022), the third design exploits the rise in Chinese import competition between 1990 and 2007, which lowered the earnings and employment of non-college workers in commuting zones whose industry mix was more exposed to imports. This is the only design in which the shock to the outside option is *negative*, and its exogeneity rests on instrumenting a CZ's exposure with the growth of Chinese exports to eight other high-income countries, which isolates the supply-driven component of China's export surge rather than any United States demand. We estimate the design as a stacked panel of two ten-year-equivalent long differences (1990–2000 and 2000 to 2006–07), with first stage

$$\Delta \ln \mathrm{ExpInc}_{r,t} = \lambda\, Z_{r,t} + \gamma_t + X'_{r,t}\pi + u_{r,t}, \tag{7}$$

and second stage

$$\Delta M_{r,t} = \theta\, \Delta \ln \mathrm{ExpInc}_{r,t} + \gamma_t + X'_{r,t}\beta + \varepsilon_{r,t}, \tag{8}$$

where $\Delta M_{r,t}$ is the change in the 18-to-24 enrollment share in CZ r over window t, $\Delta \ln \mathrm{ExpInc}_{r,t}$ is the change in the log expected income of 18-to-29-year-olds with a high-school education or less, instrumented by the other-country import-exposure measure $Z_{r,t}$, γ_t is a window fixed effect, and $X_{r,t}$ is the standard Autor et al. (2013) control set: the start-of-period manufacturing employment share, the shares of the population that are college-educated and foreign-born, the female share of employment, the share of employment in routine occupations, an occupational offshorability index, and Census-division dummies. We use the ADH population weights and cluster by state. The sample is 722 CZs across two stacked windows ($N = 1{,}444$). For both replication designs, we reproduce the published estimate in its own units before re-scaling to the expected-income coefficient, as a check that the pipeline is doing what it should; both benchmarks are reported in Appendix C.

Flow versus stock outcomes. The three designs do not all measure college-going the same way, and the difference matters for how their estimates compare. Two of them, fracking and the China shock, use an enrollment *flow*: the share of young people enrolled in college

at a point in time. This is the object the theory speaks to, because the outside option enters the decision at the moment a person chooses whether to enroll, and a flow registers that choice directly. The housing-boom design instead uses an attainment *stock*: the share of 18-to-25-year-olds who have ever attended college. This is dictated by the design we replicate. Charles et al. (2018) summarize the whole 2000s boom in a single number per metro area, the size of its price break, and estimate one long difference, comparing attainment before the boom with attainment after. A long difference over the entire boom naturally speaks to a stock, how many more young adults ended up with any college once it was over, rather than to how many were enrolled in any one year. We keep their outcome so that we can reproduce their published estimate exactly before re-scaling the design to the outside option. Because the theory speaks to the flow, we then read this stock estimate as a lower bound, for the reason we turn to next.

A stock responds to a shock more weakly than a flow does, and the reason is mechanical. Attainment builds up across many birth cohorts, and once someone has attended college it stays on their record permanently. A shock that arrives in, say, 2005 can only change the decisions of the cohorts still choosing after 2005; everyone already past that point is unaffected. So the same shock that visibly bends a single year's enrollment flow barely moves a stock made up of a decade of cohorts. This is why the housing-boom estimate (-0.129) is smaller than the two flow estimates (fracking -0.190, China -0.171), and why we treat it as a floor rather than a competing central value. The spread across designs is thus partly a matter of what is being measured, which is one reason the calibration uses a range rather than a single number. A second difference points the same way: the housing outcome counts only residents born in the state, which mutes migration, whereas the fracking panel includes the young non-college workers who moved into booming areas. Appendix C puts all three designs on a common residency basis; the estimates barely move, and the stock still lies below both flows, so the reading survives (Table 18).

Estimates. Table 4 reports the coefficient θ from the three designs on a common scale. In each, θ is the change in the college attendance rate, measured as a share, per one-log-point change in non-college expected income, so multiplying by 0.01 gives the percentage-point response to a one-percent rise in the outside option. The three estimates lie within a narrow range despite exploiting different shocks, geographies, and eras.[3] The fracking design gives a preferred estimate of -0.190: a one-percent rise in the outside option lowers the 18-to-24 enrollment share by about 0.19 percentage points, an elasticity of roughly -0.50 at its

[3]The China-shock design's second window overlaps the housing-boom window, and the two instruments are modestly negatively correlated across the commuting zones they share, so the two should not be read as independent confirmations there; the fracking design is independent of both (Appendix C).

baseline enrollment rate of 0.383. The China-shock design gives −0.171, an elasticity of about −0.53 at its own baseline share of 0.326, close to the fracking valuedespite arriving through a negative shock in an earlier era. The housing-boom design gives −0.129. All three rest on first stages that clear conventional strength thresholds, and all three are statistically distinguishable from zero.

Table 4: The enrollment response to the non-college outside option, three designs

	(1) Housing boom (Charles et al., 2018)	(2) Fracking boom (Cascio and Narayan, 2022)	(3) China shock (Autor et al., 2013)
θ: 2SLS coefficient	-0.129^{***}	-0.190^{**}	-0.171^{***}
	(0.044)	(0.084)	(0.046)
Outside-option shock	Positive	Positive	Negative
Identifying variation	Housing demand	Shale reserves	Import competition
Instrument	Structural break	Reserves × post	Other-country expo.
Level of variation	Metro area	Commuting zone	Commuting zone
Outcome	Δ any-college share (18–25)	Undergrad. enroll. share (18–24)	Δ enroll. share (18–24)
Design	Long diff. 00–07	Panel, 03–15	Stacked diff. 90–07
First-stage F	14.9	21.8	28.9
Observations	275	3,110	1,444
Clusters	48 (state)	311 (CZ)	48 (state)

Notes. Each column reports the same object: the two-stage least-squares coefficient θ of a college-attendance rate (a share) on the log of non-college expected income, instrumented as described. The designs recreate Charles et al. (2018), Cascio and Narayan (2022), and the China-shock studies (Autor et al., 2013; Conlisk et al., 2022; Ferrière et al., 2022), harmonized to a common outcome and endogenous variable (Section 2.5). Standard errors clustered as indicated, in parentheses. A coefficient θ implies a $|0.01\,\theta| \times 100$ percentage-point change in the attendance share per one-percent change in expected income; elasticities use each design's own baseline share (0.383 fracking, 0.326 China). Columns (1) and (3) carry their source studies' controls and population weights; column (2) uses CZ and state-by-year fixed effects with year-2000 CZ population weights, which is our own specification rather than Cascio and Narayan's. Full results are in Appendix C. The fracking column is the preferred boom window excluding 2005. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Tests and robustness. Several checks support reading these coefficients as the effect of the outside option on enrollment. The supporting tables and figures are in Appendix C.

The main credibility check for the fracking design is an event study (Figure 14). Before the boom, exposed and unexposed zones move together: the expected-income series is flat across 2003 to 2006, and enrollment shows no separation in the years it is observed. After 2008 the first stage climbs steadily, reaching about +12.5 log points of expected income per standard deviation of exposure by 2014, and only then does enrollment turn down, reaching about −2 percentage points by 2015. The flat pre-period in expected income is the design's central identifying evidence. The enrollment pre-period is thinner and we do not lean on it:

the ACS provides no sub-state geography before 2005, and 2005 is dropped for the group-quarters reason given below, so relative to the 2007 reference year the preferred specification has a single estimated pre-boom enrollment coefficient, in 2006.[4]

The estimate is stable across choices of the boom window. Lengthening the pre-period or including 2005 keeps the coefficient inside $[-0.16, -0.23]$, and our preferred specification sits toward the conservative end of that interval. We drop 2005 for a specific data reason. The 2005 ACS did not sample group quarters, so students living in dormitories are mechanically missing from that year's enrollment counts, and unevenly across zones. The housing design retains 2005 because its education sample excludes group-quarters residents in every year, so the omission is not a discontinuity there (Appendix B). Correcting for this moves the coefficient only from -0.164 to -0.190. Trimming barely-exposed counties, dropping the earliest-booming plays, and letting each play have its own boom timing all leave the estimate near -0.19 to -0.22.

The housing design offers less to plot, because it is a single long difference rather than an annual panel: there is no year-by-year path in which to look for a pre-trend. Its credibility rests instead on three things. The instrument is the one from Charles et al. (2018), whose case that a sharp price break reflects speculation rather than fundamentals is argued at length in their paper. Its first stage is strong, with an F of about 15 (Table 4). And its published benchmark reproduces exactly on the authors' own sample.

The China-shock design comes with its own checks (Appendix C), and it also supplies the only female-specific estimate in the range. Recall that this design is a stacked panel: it pools two long differences, 1990 to 2000 and 2000 to 2006–07, into a single regression with a window fixed effect. Its identification comes entirely from that pooling. On its own, each of the two windows has a first stage too weak to use; only when the two are stacked does a usable coefficient appear, which is why the single-window numbers are reported merely to document the point. A timing placebo, which regresses past enrollment changes on the future instrument, finds no pre-trend in enrollment and only a slight opposite-signed drift in expected income; bounding the estimate for that drift moves it to -0.142, still inside the range. The China shock also moves young low-education women's expected income strongly, which fracking does not, so it can deliver a female estimate: a coefficient of -0.206 for women against an insignificant -0.094 for men. The confidence intervals overlap, so we cannot reject that men and women respond equally. The design's real value, though, is its sign. It is the one negative shock to the outside option, in a different era and geography, and it lands right

[4] That coefficient is $+0.011$ (SE 0.008), the largest pre-period deviation in the enrollment series and of the opposite sign to the post-boom response, so it does not suggest a downward pre-trend. The longer window, which adds 2000 to 2002, is reported in Appendix C.

next to the fracking estimate. The enrollment response is therefore close to symmetric in the sign of the shock, the property the model's taste-shock scale builds in.

3 Illustrative Model

This section develops a stripped-down static model that isolates the paper's central mechanism: gender differences in non-college outside options shape the college matriculation decision, and through it the gender gaps in enrollment, student debt, and earnings. The model is deliberately minimal, with no human-capital dynamics, no college-quality choice, and no borrowing frictions, so that the comparative statics can be read off in closed form. The quantitative model of Section 4 adds these ingredients back.

Environment. The economy is static and populated by a unit mass of individuals who differ by gender $g \in \{m, w\}$ and by an idiosyncratic index $\varphi > 0$ of the financial support available to pay for college, distributed Pareto with scale χ and shape ξ. Utility is increasing in consumption, $u(c)$, and each individual either attends college or enters the labor market with a high-school degree. Women face gender wedges in the two labor markets, earning a fraction $(1 - \tau_h)$ of the high-school wage and $(1 - \tau_c)$ of the college wage; men face no wedges by normalization, $\tau_h^m = \tau_c^m = 0$. Both wedges are the empirical counterpart of the gender earnings gaps documented in Section 2, and the key feature of the model is that they may differ across the two markets. These are the same two objects the quantitative model of Section 4 writes $\tau^{f,H}$ and $\tau^{f,C}$; we use the lighter notation τ_h and τ_c here because this section has only one gender facing a wedge.

College and high-school values. A high-school worker consumes their after-wedge earnings,

$$V_g^h = \max_c \; u(c) \quad \text{s.t.} \quad c \leq (1 - \tau_h)\, w_h, \tag{9}$$

independent of φ. A college graduate earns the college wage but must finance the cost of attendance p, of which a share $1/\varphi$ is paid out of the student's own resources (loans or family transfers),

$$V_g^c(\varphi) = \max_c \; u(c) \quad \text{s.t.} \quad c \leq (1 - \tau_c)\, w_c - \frac{p}{\varphi}. \tag{10}$$

Lower financial support (smaller φ) raises the private cost of college and lowers its value. An individual matriculates whenever $V_g^c(\varphi) > V_g^h$, which, because u is increasing, reduces to a

threshold on financial support,

$$\varphi \;\geq\; \varphi_g^* \;\equiv\; \frac{p}{(1-\tau_c)w_c - (1-\tau_h)w_h}. \tag{11}$$

The denominator is the after-wedge college earnings premium, the net return to a degree,

$$\Delta_g \;\equiv\; (1-\tau_c)\, w_c - (1-\tau_h)\, w_h. \tag{12}$$

A weaker outside option (a larger τ_h) raises Δ_g in (12), lowers the threshold, and draws more of the support distribution into college.

Matriculation and student debt. Integrating (11) against the Pareto support distribution gives the group matriculation rate and, since a matriculating student borrows the share $1/\varphi$ of cost p they cannot self-finance, the average student debt among attendees,

$$\Pi_g \;=\; \left(\chi \frac{\Delta_g}{p}\right)^{\xi}, \tag{13}$$

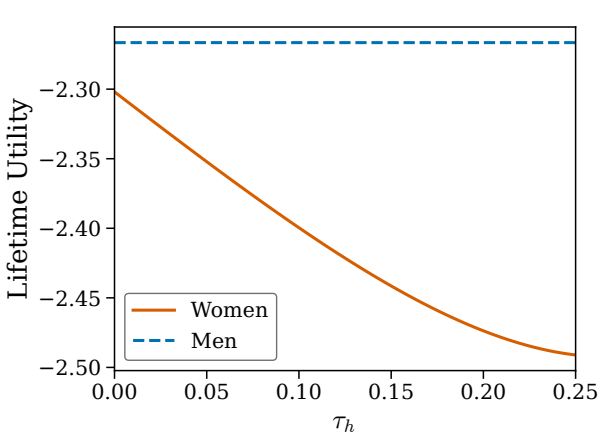
$$\mathrm{SD}_g \;=\; \mathbb{E}\left[\frac{p}{\varphi} \;\middle|\; \varphi \geq \varphi_g^*\right] \;=\; \frac{\xi}{1+\xi}\,\Delta_g. \tag{14}$$

Both objects inherit the same comparative static: a worse outside option (higher τ_h) raises Δ_g and therefore raises both matriculation and average student debt. The debt result is entirely compositional. Attendance is selected on the ability to pay, so students with more support (φ high) enroll first; a weaker outside option pulls in marginal students who have less support and therefore borrow more, raising the average. Note that the cost level p drops out of (14) entirely, even though each student borrows p/φ: because the enrollment threshold rises in proportion to p, a more expensive college selects in a richer set of students who borrow a smaller share, and under the Pareto distribution this selection exactly offsets the higher cost, leaving average debt a function of the net return Δ_g alone. This is the sense in which the model reproduces, from outside options alone, the paper's central empirical pattern that the localities and families supplying excess female matriculation also carry more debt.

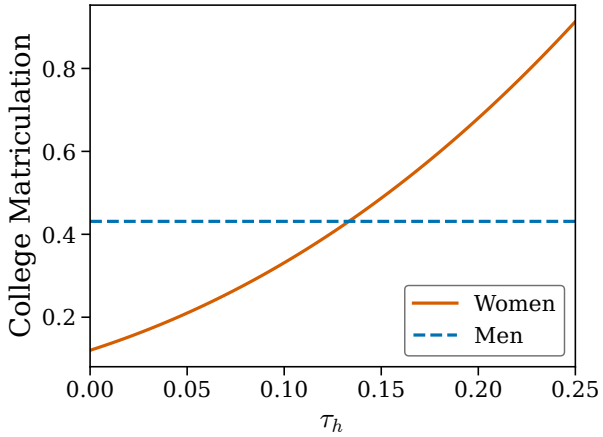
Figure 10 traces welfare, matriculation, and average debt as women's outside option worsens, that is, as τ_h rises.[5] Women's curves move monotonically and in the directions the mechanism predicts: as the outside option worsens, female welfare falls while female matriculation and average student debt both rise. Men's curves are flat throughout, which is an artifact of the static, partial-equilibrium setting: the comparative static varies only

[5] Figures 10 and 11 are a numerical illustration for a representative parameterization; they can be regenerated from the calibrated specification of Section 5.

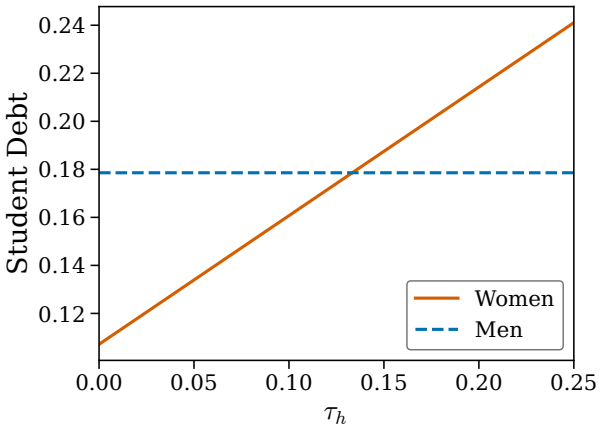
τ_h, and with the wages w_c and w_h held fixed, women's decisions cannot feed back onto men through prices. The quantitative model of Section 4 closes the labor market in general equilibrium, so that a shift in women's matriculation moves the relative supplies of college and high-school labor and, with them, men's wages and choices.



(a) Welfare



(b) Matriculation



(c) Student debt

Figure 10: Micro comparative statics as women's outside option worsens (τ_h rising). Female welfare falls while female matriculation and average student debt both rise.

The twin gender gaps. Based on the expressions above, the model's two gender gaps follow in closed form. Taking the ratio of the female to the male version of (13), and likewise of (14), and using that men face no wedges so that $\Delta_m = w_c - w_h$, gives

$$\frac{\Pi_w}{\Pi_m} = \left(\frac{(1-\tau_c)w_c - (1-\tau_h)w_h}{w_c - w_h} \right)^{\xi}, \qquad \frac{\mathrm{SD}_w}{\mathrm{SD}_m} = \frac{(1-\tau_c)w_c - (1-\tau_h)w_h}{w_c - w_h}. \tag{15}$$

The model rationalizes higher female matriculation ($\Pi_w > \Pi_m$) together with higher female debt exactly when women's net return to college exceeds men's. Since men face no wedges, $\Delta_m = w_c - w_h$, and differencing gives

$$\Delta_w - \Delta_m \;=\; \tau_h\, w_h - \tau_c\, w_c, \tag{16}$$

so that $\Delta_w > \Delta_m$ if and only if $\tau_h/\tau_c > w_c/w_h$. This is where two distinct wedges do essential work. A model with a single, common gender wedge ($\tau_c = \tau_h$) would predict that women, facing a lower return to college, matriculate at *lower* rates than men, contrary to the data. But the condition is demanding, and it is worth stating precisely what it requires: the outside-option wedge must exceed the college wedge not merely in proportional terms but by a factor larger than the college wage premium, so that the female earnings penalty is larger in *dollar* terms in the non-college market. At the wedges estimated in Section 2, $\tau_h/\tau_c \approx 1.30$ against a college premium near two, the static condition is not satisfied.[6]

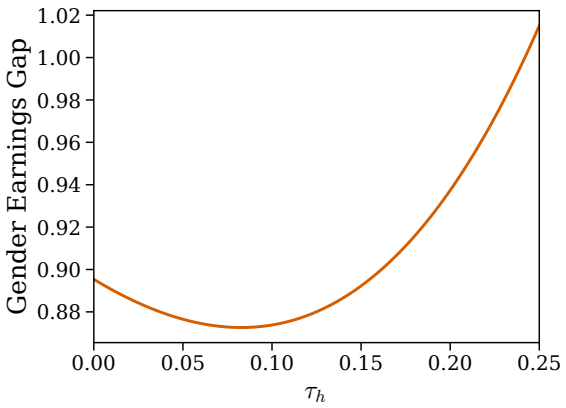
Aggregate implications. The model's purpose is to trace how these micro decisions move two headline aggregates. Summing earnings across the four gender-by-education cells gives the aggregate gender earnings gap and the aggregate college earnings premium,

$$\text{Gender earnings gap} \;=\; \frac{(1-\Pi_w)(1-\tau_h)w_h + \Pi_w(1-\tau_c)w_c}{(1-\Pi_m)\,w_h + \Pi_m\,w_c}, \tag{17}$$

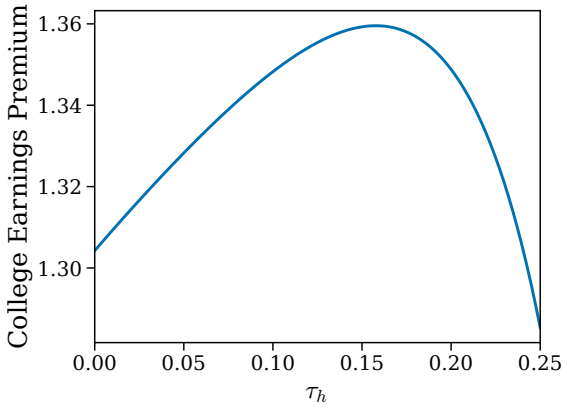
$$\text{College earnings premium} \;=\; \frac{\Pi_w^c(1-\tau_c) + \Pi_m^c}{\Pi_w^h(1-\tau_h) + \Pi_m^h} \times \frac{w_c}{w_h}, \tag{18}$$

where $\Pi_g^c \equiv \Pi_g/(\Pi_w + \Pi_m)$ and $\Pi_g^h \equiv (1-\Pi_g)/\big((1-\Pi_w) + (1-\Pi_m)\big)$ are the gender shares of the college and high-school populations. A change in women's outside option moves each aggregate through two opposing channels. Consider a worsening of women's non-college prospects, a rise in τ_h. The *direct* effect lowers the earnings of non-college women, which widens the gender earnings gap and raises the college premium. The *compositional* effect raises female matriculation Π_w, which shifts women out of the low-earning non-college pool and into college, narrowing the gender earnings gap while raising the female share of the college population and so, because women earn less within college too, lowering the college premium. Figure 11 plots both aggregates against τ_h: each is non-monotone, with the direct effect dominating at first and the compositional effect later.

[6] The comparison is not decisive against the mechanism, for two reasons the quantitative model of Section 4 makes explicit. First, this section's linear utility in the threshold comparison suppresses the curvature that makes a given proportional penalty more costly to poorer, non-college households. Second, every woman holds the outside option while the college wedge binds only on the college branch, an asymmetry in exposure



(a) Aggregate gender earnings gap



(b) Aggregate college earnings premium

Figure 11: Aggregate comparative statics as women's outside option worsens (τ_h rising). Both the gender earnings gap and the college earnings premium respond non-monotonically, and can move opposite to the direction usually taken to signal improving female welfare.

This non-monotonicity is what makes the two aggregates both interesting and treacherous to read. Each can move opposite to the direction usually taken to signal improving female welfare. The aggregate gender earnings gap can *narrow*, or even reverse sign, precisely when women's outside options worsen and their welfare falls, so it is an unreliable proxy for gender equality; likewise, the college earnings premium can shift for purely compositional reasons unrelated to the labor-market return to skill. Welfare, by contrast, is measured directly in the model by each group's expected value,

$$\text{Welfare}_g = (1 - \Pi_g)\, V_g^h + \Pi_g\, \mathbb{E}\big[V_g^c(\varphi) \;\big|\; \varphi \geq \varphi_g^*\big]\,, \tag{19}$$

and it falls unambiguously as τ_h rises even where the headline gaps improve. This divergence between the measured earnings aggregates and welfare is the illustrative model's central takeaway, and quantifying its magnitude in a realistic lifecycle setting is the task of the quantitative model in Section 4.

4 Quantitative Model

The empirical findings motivate a model in which the value of *not* attending college is a central primitive. Women enroll at higher rates than men, disproportionately in lower-income areas where non-college labor-market opportunities are weaker. They attend lower-quality, lower-cost institutions, yet they accumulate more student debt and earn less after

that the one-shot threshold here does not price. The counterfactuals in Section 6 isolate both channels.

graduation. To interpret this configuration jointly, the paper develops a stationary model of college and career choice in which worse outside options raise the net return to college, drawing more people into enrollment and into debt even when their post-college earnings remain comparatively low. Its logic runs in four steps. The choice between college and work responds to the strength of the outside option. College is not a single option but a choice of quality, and quality sets the cost of attending and hence how much a student must borrow. Post-college earnings then depend on that quality, on ability, and on gender. And the debt taken on at entry feeds forward into earnings over the rest of the lifecycle.

The illustrative model of Section 3 isolates this central comparative static: because college is chosen against the value of the outside option, a weaker outside option raises the net return to attending. The quantitative model retains that logic and adds the ingredients needed to take it to the data and, above all, to discipline the counterfactuals. The governing principle is that each added margin is itself a channel through which a change in outside options could plausibly operate, so building it in explicitly keeps the counterfactuals from crediting outside options with effects that in fact run through something else. Four such margins do the work. Endogenous college quality lets cost and debt respond to choice, so the model can speak to why women attend cheaper schools yet borrow more rather than forcing that pattern onto the wage wedge. On-the-job human-capital investment links the debt a household carries into working life to the earnings it realizes later. A production sector endogenizes the college and high-school wages, so that in counterfactuals the wage structure adjusts in equilibrium rather than aggregate gaps being attributed to composition alone. And a reduced-form college-taste shifter absorbs the non-pecuniary determinants of attendance that the structural forces leave unexplained, so they do not contaminate the mechanisms of interest. The remaining ingredients, gender-specific labor supply, initial heterogeneity in ability and parental wealth, and the extended lifecycle, are introduced as they arise below.

Notation in this section and the next is self-contained and does not carry over from Section 2, where several of the same Greek letters denote regression objects. Two conventions are worth flagging at the outset: μ denotes the money unit of \$10,000 and is distinct from μ_F, the mean vector of the initial distribution; and c denotes consumption throughout, with the cost of college written $k(q)$.

The economy is populated by a unit mass of finitely lived, forward-looking households who differ by gender, ability, and parental wealth. Households pass through a pre-college decision, two working-age periods, and a reduced-form retirement. The two working-age periods are structurally distinct: only in the first can workers invest time in accumulating human capital, so it is in that window that entry debt exerts its influence. The rest of this section develops these objects in turn, beginning with the population and preferences and

ending with the firm and equilibrium blocks.

4.1 Demographics and heterogeneity

Households live through four stages, sketched in Figure 12. In a *pre-period* (a decision taken around age 18) each household is endowed with a permanent ability index z and parental wealth a_0, and chooses whether to attend college and, conditional on attending, the quality q of the institution it attends. As a *young worker* (ages 25–44) the household chooses consumption c, end-of-period assets a', labor effort n, and a share $s \in [0, 1]$ of time devoted to on-the-job human-capital investment; student debt enters this stage as a negative initial asset position. As an *old worker* (ages 45–64) the household again chooses c, n, and a', but the investment window has closed, so $s \equiv 0$ and human capital is carried forward unchanged. From age 65 the household is retired, and its problem collapses to a reduced-form continuation value over the assets it carries into retirement, specified in (25). One model period corresponds to $T = 20$ years, and all monetary quantities are expressed in units of $\mu = \$10{,}000$ in 2018 PCE dollars.

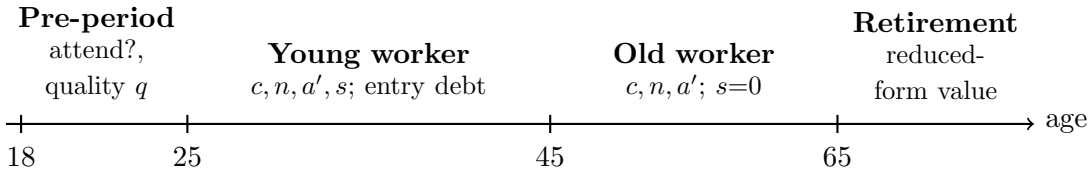


Figure 12: Model timeline.

Two attributes are fixed throughout the lifecycle: gender $g \in \{f, m\}$ and education track $e \in \{C, H\}$ (college or high school), the latter set by the pre-period choice. In addition, each household draws two state variables at birth: a permanent ability index z and initial parental wealth a_0, the resources available to help pay for college. Ability is a scalar index of innate productive capacity, normalized so that $z = 1$ corresponds to an SAT of 1000, with empirical support on roughly $[0.8, 2.4]$. For a high-school graduate, ability is directly the human-capital endowment at labor-market entry; a college graduate instead turns ability into human capital through the school quality they choose, as the human-capital subsection below makes precise. The pair (z, a_0) is drawn from a bivariate lognormal distribution F,

$$(\log z,\ \log a_0) \sim \mathcal{N}(\mu_F,\ \Sigma_F). \tag{20}$$

The covariance structure of F governs the strength of positive sorting between ability and parental resources, which determines how far initial conditions amplify the quality and ma-

triculation gaps the model generates. A single assumption disciplines the interpretation of every gender gap that follows: F is common across genders. Men and women therefore face identical distributions of ability and of parental wealth, and all gender differences in outcomes arise from the wedges and taste parameters below, never from initial conditions.

4.2 Preferences

As in Greenwood et al. (1988), flow utility is of the GHH form, identical in functional form across the lifecycle. For a young worker who devotes a share s of time to investment,

$$u^{g,e}(c, n, s) = \frac{1}{1-\rho}\left(c - \psi^{g,e}\,(1-s)\,\frac{n^{1+\nu}}{1+\nu}\right)^{1-\rho}, \tag{21}$$

and for an old worker the same expression with $s = 0$. Here c is consumption and n is labor supply (hours worked). The parameter ρ is the inverse intertemporal elasticity of substitution, which governs how strongly the household smooths consumption over time; ν is the inverse Frisch elasticity, which governs how responsive hours are to the wage; and $\psi^{g,e} > 0$ scales the disutility of labor and is allowed to differ across the four gender-education groups. The factor $(1-s)$ multiplying disutility reflects that time invested in human capital is neither paid nor a source of leisure: effective labor is $(1-s)\,n$, and it is this product that enters both earnings and the disutility term.

The paper adopts GHH deliberately, because it removes the income effect on labor supply. The intratemporal labor first-order condition then yields a closed form for hours,

$$n^{g,e}(h) = \left(\frac{(1-\tau^{g,e})\,w^e\,h}{\psi^{g,e}}\right)^{1/\nu}, \tag{22}$$

which depends on the after-tax wage and human capital alone, independent of wealth and of the investment share s. This property matters for the paper's mechanism: because hours do not respond to wealth, the effect of student debt operates entirely through the intertemporal, human-capital-investment channel and is not confounded by a labor-supply response to being in debt.

4.3 Gender wedges

Systematic gender differences in earnings that the model does not generate endogenously are carried by a *wage wedge* $\tau^{g,e}$: women earn a fraction $\left(1-\tau^{f,C}\right)$ and $\left(1-\tau^{f,H}\right)$ of the per-efficiency-unit wage in the college and high-school sectors respectively, while men face

no wedge by normalization, $\tau^{m,e} = 0$. The wedge is a reduced-form stand-in for the margins the model abstracts from, namely occupational and industry sorting within a sector, differential experience accumulation and career interruptions, the demand for temporal flexibility (Cortés and Pan, 2019), and discrimination, the channels surveyed by Goldin (2014) and Blau and Kahn (2017). The college and high-school wedges are allowed to differ because the gender earnings gap is empirically larger among the college educated. Note that the wedge shapes the outside option as well as the college payoff: a woman's high-school value is depressed by $\tau^{f,H}$, which is one of the forces raising her net return to attending. How the wedges are measured, and why they should not be read as an estimate of discrimination, is taken up in the calibration (Section 5).

4.4 Human capital

Human capital at labor-market entry differs by track. A high-school entrant carries $h = z$, with no debt. A college entrant's human capital is risky and multiplicative in school quality q,

$$h = z\,(1 + q^{\gamma})\,\varepsilon, \qquad \log \varepsilon \sim \mathcal{N}\Big(-\tfrac{\sigma^2}{2},\ \sigma^2\Big), \tag{23}$$

where $\gamma \in (0, 1)$ is the elasticity of human capital to college quality and ε is a unit-mean, log-normal post-college shock of variance σ^2, realized after enrollment and borrowing. The shock captures residual variation in post-college human capital not explained by ability and quality, such as field of study, placement, and network effects.[7] It carries more weight than a residual usually does, for two reasons: it is the model's only lever on the cross-sectional dispersion of college earnings, and, once income-driven repayment is introduced below, it is also the earnings risk that women, who earn less, insure against more often. That makes it the engine of the gender debt gap. Because ability multiplies the quality return q^{γ}, a given improvement in quality delivers a larger absolute gain to a higher-ability student, a complementarity that links the initial heterogeneity in (z, a_0) to the distribution of college choices and post-college earnings.

During the young period the worker invests a share s of time, and human capital evolves through a Ben-Porath (1967) production function with depreciation,

$$h' = B^{e}\,(h\,s)^{\alpha} + (1 - \delta)\,h, \tag{24}$$

with a track-specific scale $B^{e} \in \{B^{C}, B^{H}\}$, curvature $\alpha \in (0, 1)$, and depreciation $\delta \in (0, 1)$.

[7] It is multiplicative rather than additive so that h stays positive for every realization, and normalized to unit mean so that σ^2 identifies its dispersion and not its level.

The placement of h inside the power, as $(hs)^{\alpha}$ rather than s^{α}, is economically important: it makes the marginal return to study time increasing in h, so a worker who enters with more human capital, whether through ability or college quality, finds it more valuable to keep investing. This complementarity is how college quality comes to have lasting effects on lifetime earnings, beyond its one-time effect on human capital at entry.

The human-capital investment technology in (24) is what links entry debt to the lifecycle, a channel for which there is direct empirical support (Rothstein and Rouse, 2011; Hampole, 2024; Luo and Mongey, 2024; Alon et al., 2026). Investing s cuts the current earnings base by the factor $(1 - s)$, so its marginal cost is the foregone consumption that time would have supported, weighted by the current marginal utility of consumption. A household that enters with substantial debt ($a \ll 0$) has a high marginal utility of consumption and a strong incentive to convert income into consumption today rather than into future human capital; its opportunity cost of study time is amplified, and its optimal investment s^* is biased down relative to an otherwise identical debt-free counterpart. Heavily borrowed young women face this pressure most acutely, since the wage wedge already suppresses the return to any given stock of human capital, yielding a lifecycle earnings gap that compounds beyond the static wage gap.

4.5 Retirement

The continuation value beyond working life is a reduced-form retirement value. Following De Nardi (2004), it is a scaled CRRA value of the assets carried into retirement,

$$V^R(a') = R \cdot \frac{1}{1-\rho}\,(a' + \bar{a}_{VR})^{1-\rho}, \qquad V^R_a(a') = R\,(a' + \bar{a}_{VR})^{-\rho}, \tag{25}$$

where the scale $R > 0$ governs the strength of the old-age saving motive and the small positive shift $\bar{a}_{VR}$ keeps the marginal value of assets finite at $a' = 0$, so the saving optimum is interior and the borrowing floor never binds in old age.

4.6 The working-age decision problems

Old worker. The old-worker value function depends on the household's human capital h and financial assets a (with $a < 0$ denoting net student debt), conditional on the permanent type (g, e),

$$V^{o,g,e}(h, a) = \max_{c,\,n,\,a'} \; u^{g,e}(c, n) + \beta\, V^R(a') \tag{26}$$

$$\text{s.t.} \quad c + a' \le (1 - \tau^{g,e})\, w^e\, h\, n + (1 + r)\, a, \quad a' \ge 0,$$

where earnings are the after-tax wage times human capital times hours, β is the per-period discount factor, r is the per-period interest rate, and the continuation $V^R(a')$ is the reduced-form retirement value (25); with no labor in retirement the continuation depends only on a'. Old human capital is a fixed input over the period, and no further investment is undertaken.

Young worker. A young worker faces the same problem in either track, differing only in the education-specific prices and technology, so we write it once indexed by e,

$$V^{y,g,e}(h,a) = \max_{c,\,n,\,s,\,a'} \; u^{g,e}(c,n,s) + \beta\, V^{o,g,e}(h',a') \tag{27}$$

$$\begin{aligned} \text{s.t.} \quad & c + a' \le (1-\tau^{g,e})\, w^e\, (1-s)\, h\, n + (1+r)\, a, \\ & h' = B^e\, (hs)^{\alpha} + (1-\delta)\, h, \\ & a' \ge 0, \qquad s \in [0,1]. \end{aligned}$$

The only endogenous states are (h,a); student borrowing is captured entirely in the possibly negative initial assets carried from the pre-period, so there is no separate debt state. The human-capital investment margin is governed by,

$$\underbrace{u_s(c,n,s)}_{\text{utility of freed time}} \; - \underbrace{(1-\tau^{g,e})\, w^e\, h\, n\, u_c(c,n,s)}_{\text{value of forgone earnings}} + \underbrace{\beta\, V_h^{o,g,e}(h',a') \cdot \alpha\, B^e\, (hs)^{\alpha-1}\, h}_{\text{value of future human capital}} = 0. \tag{28}$$

The condition equates, at the margin, the direct benefit of diverting time out of paid work and the earnings that time forgoes (valued at the marginal utility of consumption) against the discounted value of the human capital the investment builds. A household carrying debt has a high u_c, which inflates the middle term and pushes s^* down: this is how student debt at entry lowers investment and hence future earnings. Single crossing of (28), guaranteed by the restriction $\alpha < \nu/(1+\nu)$, ensures a unique interior optimum.

College versus high school. The two tracks differ only in what the worker brings to (27) and in the prices and technology they face. A college graduate enters with the risky, quality-augmented human capital $h = z(1+q^{\gamma})\varepsilon$ of (23) and with entry assets $a = a_{\text{entry}}$ that may be negative when the household borrowed for tuition, and faces $(w^C, \tau^{g,C}, \psi^{g,C}, B^C)$. A high-school worker enters debt-free with $h = z$ and $a = a_0$, and faces $(w^H, \tau^{g,H}, \psi^{g,H}, B^H)$. The high-school value $V^{y,g,H}(z, a_0)$ is the household's outside option: a lower high-school wage, a larger female wage wedge, or a higher disutility lowers it and thereby raises the net return to attending college.

4.7 College cost, aid, and entry assets

The sticker cost of a college of quality q is gender-neutral and convex,

$$k(q) = p + \omega\, q^{\zeta}, \tag{29}$$

where p is a base tuition, ω scales the quality-cost gradient, and ζ is its curvature. Cost must be convex enough in quality relative to the human-capital return for the quality optimum to be interior; the calibrated value satisfies this comfortably, and the condition is stated in Appendix D. Need-based grant aid is a first-order feature of how students finance college in the United States, and it is central to the paper's question because it determines which low-wealth students can afford selective, expensive institutions and how much they must borrow to do so. The model captures the means-tested structure of the federal Pell program in reduced form, as a grant that is largest for the lowest-wealth households and phases out linearly in parental wealth,

$$\text{subsidy}(a_0) = \Phi\, \max\!\left(0,\; 1 - \frac{a_0}{a_P}\right), \tag{30}$$

where Φ is the maximum grant, received at $a_0 = 0$, and a_P is the wealth threshold above which the grant is zero. The subsidy is gender-symmetric, since aid schedules do not differ by gender. The effective, post-aid cost is

$$k_{\text{eff}}(q, a_0) = \max\big(0,\; k(q) - \text{subsidy}(a_0)\big), \tag{31}$$

with the outer max preventing the grant from making school better than free, and because the subsidy depends on a_0 only, it does not enter the quality decision. A college attender then enters the young period with initial assets

$$a_{\text{entry}} = a_0 - k_{\text{eff}}(q, a_0), \tag{32}$$

so a negative a_{entry} is student debt, while a high-school entrant simply carries a_0. The borrowing constraint $a' \geq 0$ in (27) then requires all college debt to be repaid within the young period.

4.8 Income-driven repayment

Federal student debt in the United States can be repaid on an income-driven schedule, under which a borrower pays a fixed share of income above a floor and has any remaining balance forgiven after a set period. This is insurance: it caps the downside of borrowing for households whose post-college earnings turn out low. The model builds it in as a limited-liability option, following Ji (2021). We adopt his institutional structure for income-driven repayment while abstracting from the endogenous job search that is his focus, so here the repayment insurance shapes behavior through human-capital investment and college choice rather than through reservation wages. A college household that borrowed for tuition enters the young period with negative assets a_{entry}, and it can handle that debt in one of two ways. It can repay in full, as in the base problem (27), carrying a_{entry} into working life; or it can take the income-driven option, paying only

$$\varsigma \max\big(y - \bar{y},\ 0\big), \tag{33}$$

a fixed share ς of young-period income y above a floor $\bar{y}$, with any balance it does not clear this way forgiven at the transition to old age and no asset test. The household takes whichever is better,

$$V^{y,g,C}(h,a) = \max\Big\{\, V^{y,g,C}_{\text{full}}(h,a),\quad V^{y,g,C}_{\text{IDR}}(h)\,\Big\}. \tag{34}$$

A household whose debt is small relative to its income simply repays in full and keeps a_{entry}; a household whose debt is large relative to its post-college income has the income-driven cap bind, and the unrepaid remainder is forgiven. The branch choice takes a threshold form: for each level of human capital there is a unique level of entry assets below which the household takes the income-driven option and above which it repays in full. Because the forgiven balance leaves the budget, the discharge branch depends on income alone and not on the size of the debt, so it is one-dimensional and cheap to solve. The two constants ς and $\bar{y}$ are set to statutory values outside the estimation and are reported with the externally fixed parameters.

Income-driven repayment has been modeled before ours, structurally by Ionescu (2009) and, outside the structural literature, by Dearden et al. (2008); our limited-liability formulation is closest to Ji (2021).

The structural literature on federal student lending has studied enrollment and default under the standard program (Ionescu, 2009) and the insurance value of loans against completion risk (Chatterjee and Ionescu, 2012); Dearden et al. (2008) analyze the distributional

incidence of England's income-contingent funding reform. Our limited-liability formulation follows Ji (2021).

Income-driven repayment bears asymmetrically on women, and this drives the gender gap in debt. Because a woman's post-college earnings are lower, through the wage wedge, the income-driven cap binds for her more often, so a larger share of any loan she takes is expected to be forgiven rather than repaid. The expected cost of a marginal borrowed dollar is therefore lower for women, and, facing a lower effective price of borrowing, they optimally take on more debt at origination even though college pays them less. At the model's optimum a much larger share of female than male borrowers reach the discharge region (17.7 against 3.7 percent). In the model the gender gap in debt thus runs through the repayment schedule rather than through the price or quality of schooling, and it is this channel that reconciles two facts that would otherwise sit in tension: women earn less after college yet borrow more to attend it.

4.9 Borrowing limit

Federal student lending is capped by statute, and the model imposes the cap directly. Entry debt cannot exceed the aggregate federal loan limit,

$$-\min\{0,\ a_{\text{entry}}\} \ \leq\ d_{\max}. \tag{35}$$

The cap does not restrict quality directly, since a household can always attend a better school by paying out of parental wealth; what it binds is the *borrowed* portion of the cost. Because a_{entry} falls in quality, the constraint is equivalent to a wealth-dependent ceiling $\bar{q}(a_0)$ on the quality a household can *finance*, the best school whose post-aid cost does not exceed $a_0+d_{\max}$. This ceiling binds only toward the bottom of the wealth distribution, and it interacts with need-based aid: at zero parental wealth, Pell keeps a moderate quality reachable despite the cap, while wealthier students, who finance cost from transfers rather than loans, are effectively unconstrained. The cap is thus the model's credit constraint on college quality, and it reinterprets, as a ceiling on financeable quality, the credit constraint on human-capital investment studied by Lochner and Monge-Naranjo (2011). A second credit constraint, that college debt be repaid within the young period, bears on the timing of consumption rather than on quality.

The cap plays a second, essential role under the income-driven repayment of the previous subsection. There, in the discharge region, the marginal borrowed dollar is effectively free: a borrower whose payment is set by income rather than by balance owes nothing extra on an additional dollar of debt, and the remainder is forgiven. Without a limit, such a

borrower would rationally borrow without bound, a pure moral hazard, so the statutory cap is what rules this out, in the model as in the real program, where statutory borrowing limits and income-driven repayment coexist by design. The cap therefore does two jobs at once: it bounds leverage for everyone, and it forecloses the borrow-without-limit incentive that income-driven repayment would otherwise create. Its calibrated value and provenance are given with the externally set parameters below.

4.10 College choice

The pre-period decision has two parts, a quality choice conditional on college attendance and a discrete attendance choice.

Intensive choice. Conditional on attending a college of quality q, and because entry human capital is risky in ε, the value of that choice from state (z, a_0) is the expectation over the graduation shock of the realized young-college value,

$$V^{C,g}(z, a_0, q) = \mathbb{E}_\varepsilon\big[\, V^{y,g,C}\big(h,\, a_{\text{entry}}\big)\big] \quad \text{s.t.} \quad h = z\,(1 + q^\gamma)\,\varepsilon, \quad a_{\text{entry}} = a_0 - k_{\text{eff}}(q, a_0), \tag{36}$$

evaluated on the feasible range $q \in [\underline{q}, \bar{q}(a_0)]$ set by the loan cap (35). The chosen quality maximizes this value on the feasible range,

$$q^{*,g}(z, a_0) = \arg\max_{q \le \bar{q}(a_0)} V^{C,g}(z, a_0, q), \qquad V^{C,g}(z, a_0) = V^{C,g}\big(z, a_0, q^{*,g}(z, a_0)\big), \tag{37}$$

so the value of attending is the value at the chosen school. Quality pays only through the pecuniary channel, raising entry human capital through γ and cost, and hence debt, through ζ and ω. The maximization is performed by direct comparison of the conditional values over the quality grid rather than by a first-order condition, because the objective is not globally single-peaked under income-driven repayment: beyond the discharge threshold the marginal value of entry assets is zero, so additional quality is costless up to the borrowing limit and the objective's derivative jumps at the threshold. The single-crossing conditions on study effort and quality curvature continue to hold within each repayment branch.[8]

Extensive choice. The household weighs the value of attending college at its optimal quality, $V^{C,g}(z, a_0)$, against the value of the high-school outside option, $V^{H,g}(z, a_0) \equiv$

[8]The model abstracts from an idiosyncratic school-match taste over the quality menu, of the kind that spreads otherwise identical students across nearby schools (Arcidiacono, 2004; Fu, 2014). Each type (z, a_0) therefore attends a single school, so the model reproduces neither the within-ability dispersion in attended quality documented by the undermatch literature (Hoxby and Avery, 2013; Dillon and Smith, 2017) nor the low wealth-quality correlation seen in student-level records. Reinstating it awaits a student-level moment, such as within-ability quality dispersion in the NLSY97, that would identify its scale.

$V^{y,g,H}(z, a_0)$, with each option perturbed by an i.i.d. Gumbel taste shock of scale σ_κ. It enrolls in college when

$$V^{C,g}(z, a_0) + \kappa^g(a_0) + \epsilon_C \;\geq\; V^{H,g}(z, a_0) + \epsilon_H, \tag{38}$$

where $\kappa^g(a_0)$ is a systematic college-taste shifter, depending on both gender and parental wealth, that captures non-pecuniary returns and norms not priced into the value functions. Integrating over the taste shocks gives the matriculation probability,

$$P_C^g(z, a_0) = \frac{\exp\big(\left[V^{C,g}(z, a_0) + \kappa^g(a_0)\right]/\sigma_\kappa\big)}{\exp\big(\left[V^{C,g}(z, a_0) + \kappa^g(a_0)\right]/\sigma_\kappa\big) + \exp\big(V^{H,g}(z, a_0)/\sigma_\kappa\big)}. \tag{39}$$

The difference $V^{C,g} - V^{H,g}$ is the model's net return to college: anything that lowers the outside option $V^{H,g}$ raises enrollment. The scale σ_κ governs how sharply matriculation responds to this net return, becoming deterministic as $\sigma_\kappa \to 0$ and insensitive to the systematic components as $\sigma_\kappa \to \infty$. The taste shifter is a gender-specific step function of parental wealth, constant within parental-wealth quintiles. Its shape across quintiles is estimated against the attendance-by-wealth profiles, and its level is pinned, within every calibration evaluation, by the requirement that each gender's aggregate attainment match the data; the calibration section reports the resulting schedules.

4.11 Firms

A representative competitive firm combines college and high-school effective labor through a CES technology,

$$Y = \left[A_C\, H_C^\eta + A_H\, H_H^\eta\right]^{1/\eta}, \qquad w^e = Y^{\,1-\eta}\, A_e\, H_e^{\eta-1}, \tag{40}$$

where H_e is the population-weighted aggregate of effective labor $(1 - s^*)h^*n^*$ among workers of track e, A_e is a productivity weight, and η is the elasticity parameter. The firm's factor-demand conditions equate each wage to the marginal product of the corresponding effective labor, which is what endogenizes the college and high-school wages: in the counterfactual exercises, a policy that shifts matriculation and the effective labor supplies H_C and H_H moves (w^C, w^H) through these conditions.

4.12 Equilibrium

Taking the interest rate r as given, a stationary equilibrium is a pair of wages (w^C, w^H) such that: households solve their pre-period, young, and old problems, (37), (27), and (26), taking prices as given; matriculation follows the logit (39); the firm's factor-demand conditions (40) hold, so each wage equals the marginal product of its effective labor; and the college and high-school labor markets clear, with the aggregate effective labor supplies H_C and H_H obtained by aggregating household policies over the stationary distribution, which is itself generated forward from the initial distribution $F(z, a_0)$. The calibration solves this system in partial equilibrium, fixing (w^C, w^H) at their calibrated values; the counterfactuals close it in general equilibrium, letting (w^C, w^H) adjust to clear the two labor markets. Appendix E records the numerical solution algorithm in enough detail to reimplement.

5 Calibration

The goal of the quantitative model is to assess the role of gender differences in non-college outside options in shaping college enrollment, student debt, and post-college inequality, and to trace how changes in those outside options move female welfare, the college earnings premium, and the aggregate gender earnings gap. Credibly doing so requires getting three things right. The first is the size and gender composition of student debt, which requires matching the realized distribution of debt across households while respecting that debt-holders are not a random sample but are selected endogenously through the college matriculation choice. The second is how a college education maps into earnings, which requires matching the gender patterns in post-college earnings, hours, and chosen college quality, together with the lifecycle earnings growth over which entry debt plays out. The third is how enrollment responds to the outside option, the causal relationship at the center of the paper, which requires matching the estimated effect of the non-college outside option on matriculation. The remaining targets, the wealth distribution at entry and at retirement and the initial joint distribution of ability and parental resources, are in service of pinning these three objects down in general equilibrium.

The parameters split into an externally calibrated block, fixed from prior evidence or from stand-alone estimates, and an internally calibrated block set to match a set of empirical targets. Each internal parameter is discussed alongside the moment it is most closely tied to, though in equilibrium every parameter bears on every moment, so the mapping is joint rather than one-to-one. The two aggregate attainment rates are matched exactly; the remaining targets are matched as closely as the model allows, with the fit reported below.

Table 5: Externally calibrated parameters and pinned rates

Parameter	Symbol	Value	Source / justification
Inverse IES	ρ	2.0	Standard macro (Blundell et al., 2016)
Inverse Frisch	ν	2.0	Standard macro labor (Chetty, 2012)
Discount factor	β	$0.99^{20} \approx 0.818$	Annual 0.99, twenty-year period
Real interest rate	r	$1.03^{20} - 1 \approx 0.806$	Annual 3%, twenty-year period
Study curvature	α	0.60	Literature range (Browning et al., 1999); below bound $\nu/(1+\nu) = 2/3$
Depreciation	δ	0.367	HVY level depreciation $\approx$2.26%/yr, 20-yr period (Huggett et al., 2011)
Cost curvature	ζ	1.82	Own estimate: Scorecard log-log cost-on-SAT slope, pooled 2010–2022, enrollment-weighted 4-year institutions
Quality grid	$[\underline{q}, q_{\max}]$	[0.6, 2.4]	SAT-scale range, $q = \text{SAT}/1000$
Pell maximum	Φ	2.4 μ	4× avg. real 2018$ max award, 2010–2027
IDR repayment share	ς	0.10	PAYE/REPAYE marginal rate
IDR income floor	$\bar{y}$	36.42 μ	1.5× single-person poverty guideline (Ji, 2021)
Federal loan limit	$d_{\max}$	3.1 μ	Aggregate Direct Loan limit, dependent undergrad. (FSA Handbook)
CES labor elasticity	η	0.333	Jones (2014); firm-side only, no calibration effect
Graduation-shock nodes	n_ε	7	Gauss-Hermite quadrature
Borrowing floor	$\bar{a}_{\text{borrow}}$	0	No end-of-period debt; college debt repaid when young
V^R shift	$\bar{a}_{VR}$	0.01 μ	Numerical Inada regularizer

5.1 Externally calibrated parameters

Table 5 collects the parameters fixed outside the estimation, grouped here in the order the model introduces them.

The study curvature $\alpha = 0.60$ sits in the lower-middle of the wide and weakly identified range of Ben-Porath investment curvatures, roughly 0.5 to 0.9, surveyed by Browning et al. (1999), close to the recent non-college estimate of about 0.64 in Gao et al. (2026) and just below the 0.70 of Huggett et al. (2011). It also sits safely below the single-crossing bound $\alpha < \nu/(1+\nu) = 2/3$ that the GHH labor block imposes on the study-effort first-order condition, which is the binding consideration: values in the upper half of the literature's range would break uniqueness of the study optimum.

The cost curvature $\zeta = 1.82$ is our own hedonic estimate rather than a literature value: it is the enrollment-weighted slope of a log-log regression of cost of attendance on SAT across four-year Scorecard institutions (SE 0.02, $N = 17{,}302$). It governs how steeply the price of college rises with quality, and therefore how much a student must borrow to trade up, so it does real work in the model rather than merely calibrating a level.

5.2 Data and targets

All targets are constructed on a common post-recession window, 2010–2022, deflated to 2018 PCE dollars, with each dollar quantity in model units of $\mu = \$10{,}000$. Pooling one window

rather than mixing survey vintages centers every moment on the same economic period, so the targets describe a single coherent cross-section rather than a composite of different years. This matters most for the SCF, whose single-year samples are small, but we pool every source alike so that all moments describe the same window. The blocks below say what each target disciplines and why; Appendix D gives the construction of every one of them, including sample definitions, variable choices, and the sensitivity of the two estimated slopes to those choices.

College attainment. The aggregate college rates are BA+ attainment at ages 25–30 in the pooled CPS, 0.3246 for men and 0.3961 for women, rather than enrollment rates. Attainment is the model-consistent object because the model's college decision is one-shot attend-and-complete, so completed degrees, not enrollment spells, are what the matriculation probability describes.

Student debt. Debt targets come from the National Postsecondary Student Aid Study (NPSAS), pooled across the 2012 and 2016 waves: mean debt of $0.9370\,\mu$ for men and $1.1589\,\mu$ for women, a gap of $0.2219\,\mu$. The concept is cumulative undergraduate borrowing over *all* undergraduates, including the many who borrow nothing, which is the model-consistent object: the model's mean debt averages $-\min\{0, a_{\text{entry}}\}$ over all college entrants, so its empirical counterpart must include them too. We target the gender gap directly, alongside the two levels, as the paper's central object, and the calibration matches all three closely: $0.963\,\mu$ for men (+2.8%), $1.176\,\mu$ for women (+1.4%), and a gap of $0.213\,\mu$ (−4.2%). The male level is disciplined primarily by the base tuition $p = 1.11\,\mu$ and the quality-cost scale $\omega = 3.27$; the aid phase-out threshold $a_P = 1.22\,\mu$ is pinned separately by the mean Pell award ($0.617\,\mu$).

What matters most is how the *gap* is produced. The entry cost of college is gender-neutral, so the gap is not imposed through a cost wedge. It emerges structurally from income-driven repayment: women, who earn less, reach the repayment cap far more often, and so face a lower effective price of borrowing and take on more debt at origination. Indeed the female net-cost wedge is available in the specification but is set to zero rather than estimated, so the gap is generated with no gender-specific borrowing-cost parameter at all. We read this as overidentification evidence for the repayment mechanism.

Labor-market moments. Mean earnings by track and gender, relative annual hours by cell, education-specific lifecycle earnings growth, and the variance of log earnings among young college workers are computed on the pooled 2010–2022 CPS using the full-time, full-year concept, the closest defensible counterpart to a model in which everyone with positive human capital works a full period. Male earnings by track discipline the two per-efficiency-unit wages, estimated at $w^C = 129.6$ and $w^H = 122.7$; the within-track gender earnings ratios

(0.6967 in college, 0.7307 in high school) discipline the female wage wedges, $\tau^{f,C} = 0.296$ and $\tau^{f,H} = 0.239$; relative hours discipline the four labor disutilities; lifecycle earnings growth disciplines the two human-capital scales, $B^C = 1.36$ and $B^H = 0.94$; and the variance of log college earnings (0.3948) disciplines the graduation-shock variance, $\sigma^2 = 0.049$.

The wage wedges are calibrated unconditionally: $\tau^{f,e}$ is identified by the raw full-time full-year gender earnings ratio within each sector, so it absorbs whatever portion of the observed gap the model's own margins, education choice, hours among full-time workers, and the ability composition of each sector, do not produce. Two implications follow. First, because the targets condition on full-time full-year work, the participation and part-time margins are excluded from the measured gap, so the wedge carries wage-rate differences among full-time workers rather than differences in labor supply. Second, because the remaining endogenous margins are modest within full-time cells, the calibrated wedges are close to the raw gaps. They are therefore larger than the unexplained residual of decomposition exercises, which conditions on far richer observables, and should not be read as an estimate of discrimination alone.

Wealth at entry and at retirement. Entry wealth is measured in the SCF singles sample at ages 25–30, using financial wealth net of student loans; its mean and standard deviation (1.710 and 15.20 μ) discipline the initial distribution of parental wealth. Singles are used for two reasons: single-headed households give a clean read on the head's gender, and the restriction makes this sample identical to the one underlying the attendance-by-quintile profiles below, so the wealth distribution and the attendance profiles describe the same population. Retirement wealth is the mean of the same concept at ages 65–70, measured per adult, since about two-thirds of retirees are married and a household-level measure would roughly double-count per-person wealth and misstate the saving motive. The per-adult target (42.94 μ) disciplines the retirement scale, $R = 0.386$.

Attendance by wealth quintile. Matriculation by gender across five bins of parental resources is built from the NLSY97, ranking respondents by parental net worth measured in 1997, computing BA completion by age 30 within each quintile, and then rescaling each gender's profile so that its equal-weight quintile mean equals that gender's CPS attainment aggregate. The rescaling is what makes the profiles comparable to the model's equal-mass bins, and it is why the levels below differ from the raw NLSY97 completion rates plotted in Section 2.3. The NLSY97 is preferred to the SCF here on three grounds: parental net worth is predetermined with respect to the youth's own college decision, so the ranking variable is not contaminated by the student debt that decision generates; the panel observes parental wealth, gender, and attainment together without the coresidence selection that distorts the SCF male profile; and it removes the reverse-causality concern that student

debt itself pushes a young graduate down an own-asset ranking. The resulting rescaled profiles, men $(0.092, 0.158, 0.258, 0.446, 0.669)$ and women $(0.173, 0.214, 0.355, 0.510, 0.729)$, discipline the shape of the two taste schedules across quintiles; the level of each is pinned by the aggregate attainment rate. Both are concave in parental wealth and plateau at the top. The calibration matches the low quintiles well but misses the shape higher up, most sharply for men; this is taken up in the discussion of fit.

The matriculation elasticity. The taste-shock scale is disciplined by the semielasticity of matriculation to the non-college outside option, our fracking-based commuting-zone 2SLS estimate of -0.190 from Section 2.5. Disciplining a structural taste parameter with a causally identified, partial-equilibrium elasticity is a deliberate strength of the calibration: rather than letting σ_κ absorb whatever is needed to reproduce aggregate enrollment, we pin the central behavioral margin to a quasi-experimental estimate and let the rest of the model accommodate it. The two other designs of Section 2.5 bracket this value and provide an out-of-sample check on it.

Putting the estimate and the model on the same footing takes one conversion, which is worth stating in the text because it is a substantive feature of the GHH block rather than a units convention. The data object is the enrollment response per log point of *expected income*, which is earnings times an employment rate. The model instead perturbs the wage w^H. Under GHH there is no income effect, so a one-log-point wage change raises earnings by $1 + 1/\nu = 1.5$ log points: half the response runs through the Frisch hours margin, which plays the role the employment rate plays in the data. Dividing the model's per-log-wage response by 1.5 therefore puts the two in the same units, and the constant elasticity makes the conversion exact. The estimated scale is $\sigma_\kappa = 0.0426$, at which the model semielasticity is -0.195, within about 3% of the target, and the empirical bracket $[-0.13, -0.23]$ leaves ample room around the point.

College quality. Two Scorecard moments discipline the quality block. The cross-institution slope of log earnings on log SAT (1.212) sets the earnings return to quality, $\gamma = 0.064$, which the model reproduces to within about 5%. The gender gap in attended school quality sets the quality-cost scale, $\omega = 3.27$: measured as an enrollment-weighted female-to-male ratio of institutional average SAT, women attend institutions with average SAT about 1.2 percent below men's. This is the quality of the institution attended, not students' own test scores. As external validation, the same gap built from student-level California FAFSA records gives a ratio of 0.985 against the Scorecard's 0.988, despite differing in coverage, unit of observation, and weighting, so the target is not an artifact of the Scorecard's construction.

Initial distribution. The remaining anchors are measured directly: the mean and variance of ability from the NCES SAT distribution ($z = \text{SAT}/1000$, mean 1.028, variance

Table 6: Internally calibrated parameters and model fit

Parameter	Associated moment	Target	Model	Dev. (%)
$w^C = 129.63$	Mean earnings, college male (μ)	217.12	231.22	+6.5
$w^H = 122.66$	Mean earnings, HS male (μ)	105.32	103.25	−2.0
$\tau^{f,C} = 0.296$	Gender earnings ratio, college	0.6967	0.7343	+5.4
$\tau^{f,H} = 0.239$	Gender earnings ratio, HS	0.7307	0.7351	+0.6
$\psi^{m,C} = 267.0$	Rel. hours m, C	1.0339	0.9592	−7.2
$\psi^{f,C} = 160.6$	Rel. hours f, C	0.9840	1.0246	+4.1
$\psi^{f,H} = 89.1$	Rel. hours f, H	0.9472	0.9581	+1.2
$\psi^{m,H} = 106.7$	Hours normalization ($m, H = 1$)	1.000	1.000	n/a
$B^C = 1.357$	Lifecycle growth, college	1.2653	1.2602	−0.4
$B^H = 0.939$	Lifecycle growth, HS	1.1471	1.0866	−5.3
$\sigma^2 = 0.0491$	Var(log earn), young college	0.3948	0.3502	−11.3
$\gamma = 0.0638$	q-earnings slope (indirect inference)	1.2120	1.1505	−5.1
$\sigma_\kappa = 0.0426$	Matric semielast., expected income (fracking IV)	−0.1900	−0.1950	−2.6
$\kappa^m(a_0) = [-0.118, -0.093, -0.069, -0.043, +0.054]$	Matric by quintile, male (Q1–Q5)	see Table 7; aggregate rate 0.3246 exact		
$\kappa^f(a_0) = [-0.100, -0.083, -0.049, -0.012, +0.012]$	Matric by quintile, female (Q1–Q5)	see Table 7; aggregate rate 0.3961 exact		
$p = 1.114$	Mean debt, male (μ)	0.9370	0.9630	+2.8
$\omega = 3.271$	Mean debt, female (μ)	1.1589	1.1755	+1.4
$a_P = 1.224$	Mean Pell award (μ)	0.6170	0.6200	+0.5
$R = 0.386$	Asset mean 65–70, per adult (μ)	42.94	44.70	+4.1
$\mu_F[0] = -0.0930$	Mean z (NCES SAT)	1.028	0.932	−9.3
$\mu_F[1] = -0.6062$	Mean a_{entry}, singles 25–30 (μ)	1.710	1.893	+10.7
$\Sigma_F[1,1] = 3.393$	Std a_{entry} (μ)	15.20	14.40	−5.3
$\Sigma_F[0,0] = 0.0461$ (pinned)	Var(log z), NCES SAT	0.0461	pinned	—
$\Sigma_F[0,1]$ (pinned)	Corr(log z, log a_0), NLSY97	0.3648	pinned	—

Notes. The moment shown for each parameter is the one it is most closely tied to, though in equilibrium all parameters bear on all moments. The depreciation rate δ and the income-driven-repayment constants are set externally (Table 5); $\Sigma_F[0,0]$ and $\Sigma_F[0,1]$ are pinned to the measured SAT dispersion and the NLSY97 ability-wealth correlation; and the female net-cost wedge τ_s^f and the school-match taste scale σ_q are both fixed at zero. The taste schedules $\kappa^g(a_0)$ are step functions over parental-wealth quintiles, and the two aggregate attainment rates are matched exactly. The larger residuals (the male attendance profile of Table 7 and the variance of log college earnings) are structural and are diagnosed in the discussion of fit.

of $\log z$ of 0.0461), and the ability-parental-wealth correlation (0.365) from the NLSY97, between parental net worth and a standardized cognitive-test score. The correlation is essentially equal across genders (0.366 each), which is what supports the gender-symmetric initial distribution $F(z, a_0)$ assumed above.

5.3 Internal parameters and fit

Table 6 reports the internal vector with, for each parameter, the moment it is most closely tied to and the corresponding data target, model value, and proportional deviation; Tables 7 and 8 report the quintile block and the remaining targeted and untargeted moments.

Overall fit. The calibration matches the objects at the center of the paper well. The debt block, the paper's central object, is fit on grids where it is numerically converged (Table 19) and lands inside 3% on both levels and at −4.2% on the gap. The gender

Table 7: Matriculation by parental-wealth quintile and gender (NLSY97)

Moment	Target	Model	Dev. (%)
Matric Q1, male	0.0922	0.095	+3
Matric Q2, male	0.1579	0.152	−4
Matric Q3, male	0.2575	0.234	−9
Matric Q4, male	0.4460	0.349	−22
Matric Q5, male	0.6693	0.827	+24
Matric Q1, female	0.1730	0.171	−1
Matric Q2, female	0.2140	0.217	+2
Matric Q3, female	0.3547	0.372	+5
Matric Q4, female	0.5099	0.568	+11
Matric Q5, female	0.7289	0.662	−9

Notes. Targets are NLSY97 quintiles of parental net worth (1997), BA completion by age 30, rescaled to the CPS attainment aggregates. The model reproduces the low quintiles but misses the concave shape higher up, most sharply for men, whose attendance is understated in the upper-middle quintiles and overstated at the very top. This is taken up in the discussion of fit.

Table 8: Additional targeted moments

Moment		Target/Data	Model	Dev. (%)
Mean debt gap, F−M (μ)	(targeted)	0.2219	0.2125	−4.2
SAT quality gap (f/m)	(untargeted)	0.9877	0.9392	−4.9

Notes. The SAT quality gap is the enrollment-weighted female-to-male ratio of attended-institution SAT among college entrants.

earnings ratios, college lifecycle growth, the fracking matriculation semielasticity, the mean Pell award, and the quality-earnings slope are all within about 5.5%; the two attainment rates and all external pins are matched exactly, and the overall weighted distance is small. Three groups of residuals survive, and, as sweeps at the optimum confirm, each is a structural consequence of the specification rather than slack the optimizer failed to take up.

5.4 Discussion of fit

The male attendance-by-wealth profile (the dominant miss). The data male attendance-by-wealth profile is concave and plateaus across quintiles (0.092, 0.158, 0.258, 0.446, 0.669); the model's is convex and saturates at the top (0.095, 0.152, 0.234, 0.349, 0.827). The top-quintile overshoot is +24%, and this remains the dominant and structural miss. Two levers could reshape it, and each breaks a higher-priority target. Lowering the top-quintile taste shifter does pull the richest men out, but the model must still match the aggregate male attendance rate, so pulling rich men out forces poorer men in to hold the average, and poorer men borrow more. Mean male debt then rises by about 29% and the gender debt gap flips sign; the debt block outweighs the profile. Raising the taste-shock scale σ_κ smooths the logit toward the concave data shape, but σ_κ is pinned small by the fracking instrument, and raising it enough to reshape the profile cuts the matriculation semielasticity to about -0.12, roughly 37% off target. The root cause is that σ_κ is over-identified: the instrument wants it small for a sharp matriculation response, while the curvature of the wealth-attendance profile wants it large. Economically, the model has no wealth-orthogonal reason for a rich man to decline college, so the only parameters that could produce one are already committed elsewhere; a genuine fix is a new channel, for instance an ability- or outside-option-driven reason for high-wealth, lower-return men to skip college, which is a model extension rather than a re-parameterization.

The variance of college earnings: the income-driven-repayment coupling. The earnings-shock variance σ^2 is the only lever on cross-sectional earnings dispersion, but under income-driven repayment it is also the primary engine of the debt gap, since more earnings risk means more women reach the discharge cap and the gap widens. The σ^2 that rationalizes the observed debt gap implies less dispersion than the data show; raising σ^2 to fit dispersion widens the debt gap from 0.21 to about 0.32 and sharply reduces the quality-earnings slope. The undershoot is therefore a property of the mechanism, which disciplines earnings risk to the quantity that produces the gender debt gap through discharge. Closing it would require generating part of the gap through a second channel (for instance, reinstating the female cost wedge, pinned to exactly zero) or adding a dispersion channel that does not also trigger

discharge.

Secondary misses. The female-to-male quality ratio, now left untargeted, is overstated (model 0.939 against about 0.988): the model sorts women into lower-quality schools more sharply than the data, and no parameter is dedicated to this ratio. Mean college-male earnings (+6.5%), high-school lifecycle growth (−5.3%), mean ability (−9.3%, now a clean location miss after the ability-grid correction described in the appendix), the mean (+10.7%) and dispersion (−5.3%) of entry wealth, and old-age assets (+4.1%) are lower-weight level and initial-distribution margins that never fully settle.

6 Decomposing the Gaps

The calibrated model reproduces the twin gender gaps in matriculation and student debt as equilibrium outcomes of a small set of primitives. To learn what generates them, we switch off, one at a time, the forces the calibration identifies, and read the twin gaps together with the two aggregates the illustrative model warned were unreliable welfare proxies, the gender earnings gap and the college earnings premium. This section decomposes the gaps into the two female earnings wedges the model uses to match the labor market, which are the model's only gender-specific prices and therefore the natural place to start.

The calibration matches the within-sector gender earnings ratios with two female wedges, a college wedge $\tau^{f,C} = 0.296$ and an outside-option wedge $\tau^{f,H} = 0.239$, each a proportional reduction in the female efficiency wage relative to the male wage in the same sector; men are the $\tau = 0$ normalization. The two wedges are the model's only gender-specific prices, so isolating their role isolates how much of each gap is pay and how much is everything else, namely tastes, attendance, and the general-equilibrium response of wages. Formally, we consider four experiments: removing the outside-option wedge ($\tau^{f,H} = 0$), removing the college wedge ($\tau^{f,C} = 0$), removing both, and replacing both with their common average $\bar{\tau} = 0.262$, the female-matriculation-weighted mean of the two calibrated values. Each experiment holds the taste block and every other parameter at its calibrated value and recomputes the stationary equilibrium in general equilibrium, letting the college and high-school wages adjust to clear the CES labor market (40) at the baseline-recovered technology ($A_C = 3.399$, $A_H = 2.904$, $\eta = 0.333$). Closing the model in general equilibrium matters here because the experiments move the relative supplies of college and high-school labor, and, as the premium row below makes quantitative, the equilibrium price adjustment is large enough to reverse the sign of the measured response in one experiment and to offset most of it in another.

Table 9 reports the four moments across the baseline and the four experiments. The

Table 9: Counterfactual decomposition of the gaps into the earnings wedges

Moment	Data	Model	$\tau^{f,H}=0$	$\tau^{f,C}=0$	Both $=0$	$\tau=\bar{\tau}$
	ratios; 1.000 denotes gender parity in rows 1–3					
Matriculation gap (F/M)	1.220	1.220	0.984	1.388	1.139	1.279
Student debt gap (F/M)	1.237	1.221	1.042	1.610	1.598	1.212
Gender earnings gap (F/M)	0.751	0.781	0.893	1.099	1.239	0.809
College earnings premium (C/H)	1.968	2.186	2.216	2.356	2.371	2.187

Notes. All rows are female-to-male ratios except the college earnings premium, which is the ratio of pooled college to pooled high-school mean earnings. In the Data column the first two rows are the CPS matriculation and NPSAS debt targets; the last two are implied by the targets, combining the track-level gender earnings ratios and the full-time full-year male earnings levels with the attainment rates. Each experiment sets the indicated wedge or wedges and recomputes the stationary equilibrium in general equilibrium at the baseline-recovered technology ($A_C = 3.399$, $A_H = 2.904$, $\eta = 0.333$); the taste block, the matriculation-shock scale, the labor disutilities, and all other parameters are held at their calibrated values. The common wedge $\bar{\tau} = 0.262$ is the female-matriculation-weighted average of the two calibrated wedges ($\tau^{f,C} = 0.296$, $\tau^{f,H} = 0.239$).

discussion below takes them one at a time, since each gap responds to the wedges through a different channel.

The matriculation gap. Matriculation weighs the value of attending against the value of the outside option, so each wedge moves enrollment through the side of the comparison it touches. Removing the outside-option wedge ($\tau^{f,H} = 0$) pushes the gap below one, to 0.98: the marginal female attender in the baseline is a woman of moderate ability and modest resources for whom college is largely an escape from a penalized non-college labor market, and once her non-college work pays like a man's she stays out. Removing the college wedge instead ($\tau^{f,C} = 0$) makes the degree pay women fully and draws more women in, raising the gap to 1.39. The instructive case is removing both, where female matriculation falls relative to the baseline and the gap drops to 1.14, so at the calibrated values the two wedges jointly push women into college even though the college wedge is the larger of the two. The reason is intuitive: a wedge destroys value in proportion to how much the lost consumption hurts, and non-college women are poorer, so the outside-option wedge destroys more value per point than the college wedge does; moreover every woman holds the outside option, while the college wedge binds only on the college branch. The excess female matriculation that survives when both wedges are gone, the residual 1.14, is the part carried by the taste block, the non-pecuniary pull toward college the calibration attributes to women. Finally, replacing the two wedges with their common average raises the gap to 1.28, because averaging lowers the college wedge (from 0.296 to 0.262) and raises the outside-option wedge (from 0.239 to 0.262) at once, and both changes point the marginal woman toward campus. It is

worth noting that because the calibrated college wedge exceeds the outside-option wedge, the asymmetry between the wedges does not by itself rationalize excess female matriculation in the quantitative model; the level of the outside option does, together with tastes.

The student debt gap. In the illustrative model of Section 3 the debt gap prices selection on the ability to pay, so only the relative wedge matters; in the quantitative model it prices insurance against low post-college earnings, so the level of the female college earnings shortfall matters and the relative wedge largely does not. Mean female debt has three moving parts: who attends, what they buy, and how much of a borrowed dollar a woman expects to repay. Income-driven repayment makes the third gender-asymmetric, because college pays women less, the income-contingent cap binds for them far more often (17.7 against 3.7 percent of borrowers discharge at the baseline), a borrowed dollar is therefore cheaper for a woman, and women rationally borrow more. Each column then reads naturally. Removing the outside-option wedge closes the gap to 1.04, mostly through the first part running in reverse: the low-wealth, high-borrowing women whom the weak outside option pushed into college now leave, the remaining female attenders are better resourced and higher earning, and the discharge share falls to 11.5 percent. Removing the college wedge instead shuts down the insurance channel outright, the discharge share collapsing to 1.7 percent because college now pays women well, and yet the gap rises to 1.61: with the degree fully paid, more women attend and they buy better schools, so the gap re-emerges as genuine investment borrowing that women expect to repay. The same logic carries the both-wedges-gone column (1.60). The common wedge is the sharpest demonstration. The asymmetry between the wedges is gone, yet the debt gap sits at 1.21, essentially its baseline value, with the discharge share still at 14.4 percent, because the insurance engine needs only the level of the female college earnings shortfall, which a common 26 percent wedge still delivers.

The aggregate gender earnings gap. The aggregate female-to-male earnings ratio mixes within-sector pay with the composition of where each gender works, and the experiments separate the two. Removing the outside-option wedge lifts the ratio from 0.78 to 0.89 for the most direct reason available: roughly sixty percent of women work outside college jobs, and those women now earn what men earn, so the remaining shortfall is the still-taxed college women. Removing the college wedge instead pushes the ratio above one, to 1.10, and the reason is composition rather than pay, since women attend college more than men, so once the degree pays them fully their attainment advantage converts into an aggregate earnings advantage even while non-college women remain taxed. With both wedges gone the ratio reaches 1.24: under gender-symmetric pay the model implies that women would out-earn men in aggregate by nearly a quarter, purely because more of them hold degrees. This is the quantitative counterpart of the illustrative model's warning in Section 3, namely

that the aggregate earnings gap is a composition-laden statistic that can reverse sign without any woman being paid more than an identical man, so it is a treacherous proxy for relative welfare. The common wedge leaves the ratio at 0.81, close to the baseline, since it imposes roughly the same average tax on female earnings and merely shifts its incidence between sectors.

The college earnings premium. The premium moves for compositional and equilibrium reasons rather than because the return to skill changes; nothing in these experiments touches the technology. Removing the outside-option wedge shows both forces at work. In partial equilibrium, holding wages at their baseline values, the measured premium falls to 1.87, purely mechanically, because untaxing non-college women raises measured high-school earnings, the premium's denominator. In general equilibrium it ends slightly above the baseline, at 2.22, because women leave college, so college labor becomes scarce and its wage rises (by 5.6 percent) while high-school labor becomes abundant and its wage falls (by 6.3 percent), and this supply response outweighs the mechanical one. Removing the college wedge is the mirror image. Partial equilibrium sends the premium to 2.88 by untaxing the college mean, and the general-equilibrium inflow of college labor damps it back to 2.36. The remaining two columns land between these poles for the same reasons. The lesson mirrors that for the aggregate gap. Comparing the partial-equilibrium and general-equilibrium responses shows that the equilibrium price adjustment is not a second-order correction but the dominant term: in the outside-option experiment it reverses the sign of the measured movement, and in the college-wedge experiment it offsets about three-quarters of the mechanical one. What the premium registers in these experiments is therefore mostly a price response to shifts in attainment rather than a change in what college teaches, so the premium, too, should not be read as a welfare statistic.

7 Conclusion

This paper has argued that the value of not attending college is a central determinant of who attends, how much they borrow, and what the resulting aggregates look like. Women face a proportionally larger earnings penalty in the non-college labor market than in the college market, and, because non-college women are poorer, a penalty of given size destroys more value there. Since every woman holds the outside option while the college penalty binds only on the college branch, a weak non-college market raises the net return to a degree even where the degree itself pays comparatively little. The empirical footprint of the mechanism is visible in where excess female matriculation comes from, disproportionately low-income counties and low-wealth families, and in where it goes, disproportionately to lower-quality institutions

whose graduates earn least. Three quasi-experimental designs, harmonized to a common outcome and a common endogenous regressor, put the semielasticity of matriculation to the outside option at about -0.190, with the three headline designs spanning -0.129 to -0.190.

Disciplined by that estimate, a quantitative lifecycle model matches the twin gender gaps in matriculation and student debt, both of which are calibration targets. What the calibration does not impose is the source of the debt gap: the entry cost of college is gender-neutral and the female net-cost wedge is fixed at zero, so the gap emerges from income-driven repayment. Because college pays women less, the income-contingent cap binds for them far more often, lowering the expected price of a borrowed dollar and raising optimal borrowing at origination. The most consequential counterfactual is also the most uncomfortable: paying women exactly as identical men in both sectors would raise the aggregate female-to-male earnings ratio to 1.24, because women hold more degrees. The aggregate gender earnings gap and the college earnings premium are therefore composition-laden statistics that can reverse sign without any change in relative pay, and neither is a reliable measure of gender progress.

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A Additional Tables for the Empirical Analysis

This appendix collects the supporting tables for the empirical analysis: the summary statistics for the CPS ASEC sample underlying the gender wage-gap regressions, and the school-level regressions of the female–male debt gap that the within-school evidence draws on.

Table 10: Summary Statistics

	N	Mean	SD
Real annual wage	943,171	74,811.95	81,680.55
Female	943,171	44.59	49.71
Has at least academic associate degree	943,171	48.59	49.98
Has at least bachelor degree	943,171	42.23	49.39

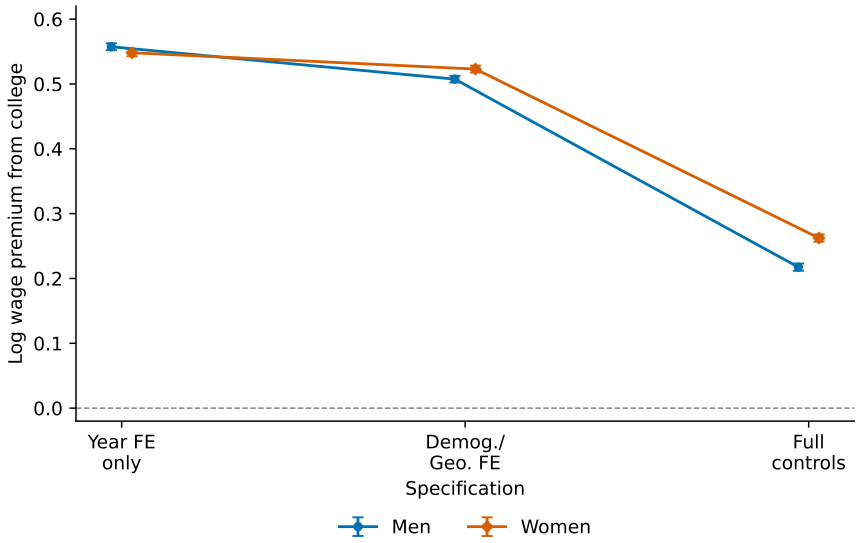
Real annual wage in 2022 dollars (CPI-U). All other variables in percentage points. Weighted (ASEC person weights). Sample: full-time workers aged 25–54 with at least a high school degree, survey years 2004–2022, positive wage income and weeks worked.

Table 11: Female–Male Median Debt Gap: OLS Regressions

	(1)	(2)	(3)	(4)	(5)
Avg SAT Score (std)	-4.192***	-5.148***	-4.915***	-2.327***	-1.668**
	(0.307)	(0.359)	(0.397)	(0.599)	(0.743)
Cost of Attendance (std)		1.952***	1.837***	-0.195	3.532***
		(0.347)	(0.364)	(1.141)	(1.063)
Female Share (std)			0.591	0.727**	-5.451***
			(0.384)	(0.369)	(0.951)
% with Federal Loan (std)				3.096***	0.593
				(0.393)	(0.422)
Admissions Rate (std)				0.392	0.233
				(0.328)	(0.267)
HBCU				1.794	
				(1.436)	
Year FE	Yes	Yes	Yes	Yes	Yes
Sector FE	No	No	No	Yes	No
School FE	No	No	No	No	Yes
Observations	13,913	13,913	13,913	13,913	13,913
R-squared	0.120	0.139	0.140	0.180	0.621
Adj. R-squared	0.119	0.138	0.139	0.179	0.579

Notes: Dependent variable is the female–male median debt gap (%). All specifications include year fixed effects. Columns (1)–(4) are cross-sectional (no school fixed effects); column (5) absorbs school fixed effects. Regressors are standardized to mean 0, SD 1 on the estimation sample. Enrollment-weighted; standard errors clustered at the school level in parentheses. Sample: four-year public and nonprofit institutions (Scorecard sectors 4 and 5, specialized and tribal Carnegie classes excluded) with non-missing debt gap and regressors. Gendered debt medians cover Scorecard cohorts with year_end 2002–2021; joint availability of SAT, cost, admissions rate, and federal-loan share restricts the estimation sample to year_end 2010–2021. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 13: College Wage Premium by Gender



Notes: Estimated college wage premiums for men ($\hat{\beta}_1$, blue) and women ($\hat{\beta}_1 + \hat{\beta}_3$, vermillion) from equation (1), plotted across three specification levels on the pooled 2004–2022 sample. In the raw specification the premiums are nearly identical (approximately 0.56 and 0.55 log points); as occupation and industry fixed effects are added, the female premium diverges upward, exceeding the male premium by approximately 4.5 log points in the full specification (0.262 against 0.217). Brackets are 95% confidence intervals. Sample: full-time workers aged 25–54, CPS ASEC 2004–2022.

B Data Appendix

This appendix collects the data construction underlying Section 2, relocated here so the main text can state the facts without the measurement apparatus.

Primary datasets

Current Population Survey. The gender earnings gaps of Section 2.2 are estimated from the Annual Social and Economic Supplement (ASEC) of the Current Population Survey, extracted from IPUMS (Flood et al., 2024). The canonical estimation sample pools survey years 2004 through 2022 and covers full-time workers aged 25 to 54 with at least a high-school diploma and positive supplement weights; wage and salary income is deflated to 2022 dollars with the CPI-U of the income year. Two education classifications are used, each for its purpose: the descriptive regressions group workers by whether they hold at least an academic associate degree (IPUMS educ 92 or higher; holders of vocational associate degrees fall in the non-college group, a threshold inherited from the original build and flagged for author review), while the wedge estimates underlying the model use a clean split, exactly a bachelor's degree against exactly a high-school diploma, so that the wedges are not diluted by associate degrees or incomplete college. Three earnings concepts appear, annual earnings, weekly wages, and hourly wages, and Table 3 reports how the implied wedges vary across them; the hourly concept is the model-facing one.

College Scorecard. The cross-college facts of Section 2.4 are built from the U.S. Department of Education's College Scorecard institution-level files (U.S. Department of Education, 2024), an unbalanced school-year panel covering academic years 2001–02 through 2022–23, matched one-to-one to IPEDS institutional characteristics and restricted to four-year public and four-year private nonprofit institutions (2,263 institutions before the exclusions below, 1,717 after). The key outcome is the female–male gap in median cumulative federal loan debt among borrowers, expressed in percent of the male median; the Scorecard reports these gendered debt medians through the 2020–21 file, so the debt-gap panel ends in 2021. Institutional covariates include the average SAT score of entering students, the annual cost of attendance (in 2022 dollars using the CPI-U), the female share of undergraduates, and median earnings of enrollees ten years after entry, which the Scorecard constructs from Treasury records for the pooled 2009–10 and 2010–11 entry cohorts. Throughout, institutions are weighted by undergraduate enrollment, and specialized and tribal institutions are excluded.

NLSY97. Evidence on matriculation by family resources comes from the National Longitudinal Survey of Youth 1997 (Bureau of Labor Statistics, U.S. Department of Labor, 2024),

which follows a cohort born 1980–84. The outcome is completion of a bachelor's degree at or before age 30, constructed from degree-date records; the resource measure is parental household net worth measured in 1997, before the college decision; and all statistics use custom longitudinal weights for the subsample observable through age 30 (final sample 5,913: 2,971 men and 2,942 women). Because parental wealth is measured pre-decision for essentially all youth regardless of later living arrangements, this design avoids both the coresidence selection and the outcome contamination that affect wealth measured in early adulthood.

Datasets for the causal designs

American Community Survey. We measure college enrollment from the American Community Survey (ACS), extracted from the IPUMS samples (Ruggles et al., 2024), combining the 2000 Census five-percent long form with the 2005–2024 ACS. The enrollment outcome is the share of residents aged 18 to 24 who report currently attending an undergraduate program, computed person-weighted within geography and time. Because the ACS is released on Public Use Microdata Areas rather than labor markets, we allocate its cells to 1990 commuting zones (CZs) using population-based crosswalks, so that enrollment lines up with the geography of the local demand shocks.

Quarterly Workforce Indicators. We build the non-college outside option from the Quarterly Workforce Indicators (QWI), the public tabulations of the Census Bureau's Longitudinal Employer-Household Dynamics program (U.S. Census Bureau, 2026). For workers with a high-school education or less, the QWI report both the average earnings of stable, full-quarter employees and total employment, from which we construct *expected income*, defined as average earnings scaled by the employment rate of the same group.

Shale reserves and play geography. The fracking design uses shale exposure from the U.S. Energy Information Administration (EIA). We take per-play proved reserves from the EIA's annual reserves reports and the boundaries of tight-oil and shale-gas plays from the EIA's May 2011 play map (U.S. Energy Information Administration, 2015). Reserves are allocated to counties by each county's share of overlapping play-polygon area, converted to common energy units, aggregated to CZs, and expressed per capita, giving a fixed, geology-based measure of exposure to the post-2007 boom.

Fracking sample definition. A major shale play is a formation for which the EIA reports reserve estimates; the seventeen states are those in which at least one commuting zone overlaps such a play on the EIA's May 2011 play map, the sample definition of Cascio and Narayan (2022). Commuting zones are assigned to the state containing the largest share of

their year-2000 population, giving 311 CZs. Unlike Cascio and Narayan, who exclude three states with limited early QWI coverage and trim extreme-size CZs (202 CZs in 14 states), we retain all seventeen states without trimming; the fringe-trim robustness in Appendix C plays the corresponding role.

Population data. County and CZ population denominators, including the year-2000 populations used to form per-capita reserves and the fixed population weights used throughout, come from the SEER U.S. population series (Surveillance, Epidemiology, and End Results (SEER) Program, 2024).

Replication packages. The housing-boom design is built on the replication package of Charles et al. (2018), which provides the metropolitan-area panel of young low-education labor-market outcomes, the college-attendance outcome, and the authors' housing-demand structural-break instrument. The China-shock design is built on the Autor et al. (2013) package, which provides commuting-zone import exposure, the other-country instrument, the ADH control set, population weights, and state clusters. We use both directly, re-scaling each design so that the endogenous variable is the outside option rather than the original shock.

Group quarters and the 2005 ACS. The 2005 ACS sampled no group quarters, so students living in college dormitories are mechanically absent from that year's enrollment counts, and unevenly across geographies. The two flow designs (fracking and China) include dormitory residents and therefore drop 2005; the housing design's education sample excludes group-quarters residents in every year, so its pooled 2005–2007 endpoint can retain it without inconsistency.

Deflators. All earnings in Section 2 are expressed in constant 2022 dollars using the annual CPI-U, consistent with the project standard. Note that the model targets of Section 5 instead use 2018 PCE dollars; the two conventions are kept separate throughout and never mixed within an exhibit.

C Causal Evidence: Additional Results

This appendix collects the supporting output for the three causal designs of Section 2.5: the fracking event study, the window-sensitivity and robustness variants of the fracking design, the gender splits, the full housing-boom and China-shock tables including their published benchmarks, and the residency harmonization. Dataset sources and sample definitions are in Appendix B; what follows is the variable construction specific to each design. All tables report the coefficient on the instrument (first stage, reduced form) or on the instrumented expected-income measure (2SLS), with standard errors clustered by CZ (fracking) or state (housing boom, China shock) in parentheses. Stars: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Variable construction

Enrollment outcomes. Each design measures enrollment differently, and each is person-weighted. The fracking outcome is current undergraduate enrollment among 18-to-24-year-olds (IPUMS `SCHOOL`= 2 and `GRADEATT`= 6), all residents including those in group quarters (college dormitories), on the ACS 2003–2015 (2005 dropped; see below). The China-shock outcome uses the same enrollment flow for 2000 and later; for the 1990 Census, which has no `GRADEATT`, it uses a proxy, enrolled with at least high-school attainment (`SCHOOL`= 2 and `EDUCD`$\geq$ 62), on 18-to-24-year-olds, all residents, excluding Alaska and Hawaii to match the ADH universe. Two rules discipline the China outcome. Within each long difference the same definition is used at both endpoints, the EDUCD proxy for the 1990–2000 window and the GRADEATT definition for the 2000–2006/07 window, so no window mixes definitions and the definitional seam is absorbed by the window fixed effect; on the 2000 Census, where both are computable, the proxy overshoots the GRADEATT definition by 1.6 percentage points (33.2 against 31.9 percent). The published-benchmark reproduction (Table 17) additionally counts graduate enrollment (`GRADEATT` 6 or 7) to match the published "enrolled in any year of college," a choice that is immaterial ($\theta = -0.171$ against -0.175). The housing outcome is CHN's any-college attainment share (completed at least one year of college), `d_any_18_25`, for 18-to-25-year-olds, taken from their replication package (2000 Census and 2007 ACS) and restricted to residents born in their state of residence, CHN's own `_same` sample.

Expected income. All three expected-income measures share one concept, a price times an employment-inclusive quantity, but differ in source and population. For fracking the source is the QWI administrative data: the price is real average monthly earnings of full-quarter stable high-school-or-less employees (QWI `EarnS`, education groups E1 and E2, employment-weighted), and the quantity is QWI high-school-or-less employment divided by

the SEER commuting-zone population aged 25 and over. For housing the measure is CHN's composition-adjusted wage of 18-to-29-year-olds with high school or less (their wage bill over their employment count), and the first stage instruments the change in that log expected income with the structural-break magnitude. For China it is per-capita annual wage income (`INCWAGE`) of 18-to-29-year-olds with high school or less, including zeros, so it is already employment-inclusive. The fracking measure therefore covers the adult non-college workforce (25+) while the two replication designs measure the young non-college group directly; the China design's not-enrolled income variant (-0.207 against -0.171) shows this does not drive the estimates.

Geography and crosswalks. Microdata are assigned to 1990 commuting zones with Dorn's population-share crosswalks (fractional `afactor` allocation): 1990 Census PUMAs through Dorn's 1990 crosswalk, 2000 Census and ACS 2005–2011 through the 2000-PUMA crosswalk, and ACS 2012 and later through geocorr (2012 and 2022 PUMA vintages). The China design assigns 1990 through PUMAs rather than counties because `COUNTYFIP` is unidentified for 37.5 percent of 1990 18-to-24-year-olds; county-to-CZ mapping uses Dorn's `cw_cty_czone` with a few FIPS recodes.

Reserves (fracking). Per-capita shale reserves interact area-allocated play reserves with year-2000 SEER population. Reserves are the maximum across EIA report editions (natural gas 2008–2015, oil 2011–2015), keeping only plays that appear in at least four editions; energy is converted at 1,022 BTU per cubic foot of gas and 5.85 MMBTU per barrel of oil; and play boundaries are the EIA May-2011 play map. Exposure is standardized within the estimation sample.

Samples and endpoints. The fracking sample is the 311 commuting zones in the seventeen shale states (state list and inclusion rule with the design in the main text); the housing sample is CHN's 275 metropolitan areas, on which the benchmark reproduces exactly; the China sample is the 722 mainland commuting zones of the ADH universe (Alaska and Hawaii dropped), with one crosswalk-unmatched cell, a Katrina-displaced Louisiana PUMA present in 2006–07 only (0.05 percent of weight). The China design's second window ends in 2006–07 rather than 2005–07 for the reason 2005 is dropped from fracking: the 2005 ACS excluded group quarters, so the endpoint pools 2006–2007 (ADH's own endpoint was ACS 2006–2008). The two window changes are converted to ten-year equivalents by the factor 10/6.5 (endpoint midpoint 2006.5), applied to both the enrollment change and $\Delta \ln \text{ExpInc}$ so the Wald ratio is scale-consistent.

Standard errors. Fracking standard errors follow the Stata `reghdfe` degrees-of-freedom convention, and the fracking chain is verified digit-for-digit across Python, R (`fixest`), and Stata (`reghdfe`/`ivreghdfe`). The housing and China designs use cluster-robust standard

errors with the small-sample correction and p-values from $t(G-1)$ with $G = 48$ states; the China design has been independently reproduced to six or more decimals, with its Stata cross-language pass still pending.

Exclusion restrictions. The bundled-channel caveat of the main text takes a design-specific form. In fracking the boom also moves household income, migration, and local public finance (including school-district revenue in shale states), so θ is the net enrollment response to the whole bundle. In housing the CHN structural-break exclusion restriction is debated in the literature and inherited as published; because parental home equity is itself a college-finance channel that works against finding enrollment declines, the -0.129 is conservative on that margin. In China the published PSID evidence attributes the response to wealthier households, direct evidence that the parental-resource channel is active alongside the opportunity-cost channel; the two have opposite signs for a negative shock, so θ is a net response whose compliers differ from the boom designs'.

Housing elasticity. The main-text table reports outside-option semielasticities for the fracking and China designs but not for housing, deliberately: the housing outcome is an attainment stock, so dividing θ by a baseline would combine a diluted numerator with a stock baseline and is not the same elasticity object; any housing elasticity would require a flow-loading adjustment rather than a simple division.

Fracking event study

Figure 14 is the credibility exhibit for the fracking design. It plots year-by-year coefficients from event-study versions of the first stage (3) and its corresponding reduced form (enrollment regressed on the same interacted exposure), replacing the single $\text{Reserves}_c \times \text{Post}_t$ term with standardized exposure interacted with year indicators relative to the 2007 reference. The expected-income first stage is flat before the boom and rises steadily thereafter, reaching about $+0.13$ log points per standard deviation of exposure by 2014 before rolling over in 2015 with the boom's production profile. The enrollment reduced form shows no pre-2008 trend and turns negative only after the boom begins, reaching about -0.02 by 2015. The absence of a pre-trend in enrollment is the design's central identifying evidence.

Fracking window sensitivity and robustness

Table 12 shows that the fracking estimate is stable across the family of boom windows. Lengthening the pre-period raises the magnitude modestly, and the preferred specification (boom window, excluding 2005) sits at the conservative end. The 2005 ACS did not sample

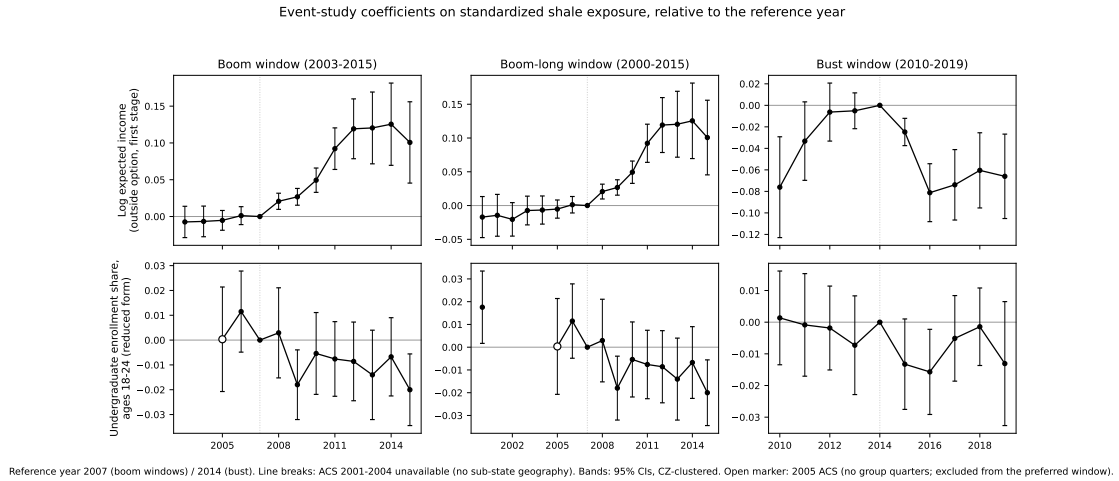


Figure 14: Fracking event study, commuting-zone level. Event-study coefficients on standardized shale exposure interacted with year indicators, relative to the reference year (2007 for the boom windows, 2014 for the bust). Columns are the boom (2003–2015), boom-long (2000–2015), and bust (2010–2019) windows. The top row is the expected-income outside option (the first stage of (3), log expected income of high-school-or-less workers); the bottom row is the enrollment reduced form (the 18-to-24 undergraduate share regressed on the interacted exposure). The preferred specification is the boom column, excluding the 2005 ACS (shown as an open marker), which sampled no group quarters. Breaks in the series reflect ACS years 2001–2004, which lack sub-state geography. Bands are 95% confidence intervals with CZ-clustered standard errors.

group quarters, so dormitory residents are mechanically excluded from that year's enrollment counts in a way that varies across CZs; dropping 2005 is a measurement choice that moves the coefficient from −0.164 to −0.190, not an identification choice. Table 13 reports the robustness block on the preferred window: allowing play-specific boom timing, dropping the earliest-booming (Barnett and Fayetteville) CZs, and zeroing marginally exposed counties all leave the estimate between about −0.19 and −0.22.

Gender split (fracking, preferred window)

The fracking shock is strongly male-tilted in the outside option, as expected, but the enrollment response is not clearly tilted by sex. Table 14 reports the first stage, reduced form, and 2SLS by sex. The male first stage is roughly 3.6 times the female first stage, yet the male and female reduced forms have overlapping confidence intervals. The female first stage is too weak (F below conventional thresholds) to support a credible female-specific 2SLS, so the female 2SLS is reported only for completeness. The calibration therefore uses the pooled coefficient rather than a male-female ratio constructed from this design.

Table 12: Fracking design: window sensitivity

Window	2SLS θ	SE	First-stage F	N
Boom (2003–15)	-0.164^{**}	(0.065)	21.8	3,421
Boom, excl. 2005 (preferred)	-0.190^{**}	(0.084)	21.8	3,110
Boom-long (2000–15)	-0.197^{***}	(0.074)	21.0	3,695
Boom-long, excl. 2005	-0.226^{**}	(0.095)	20.7	3,384

Notes. 2SLS coefficient on log expected income from (4), by boom window. CZ fixed effects and state-by-year fixed effects throughout; weighted by year-2000 CZ population 18–24; standard errors clustered by CZ (311 clusters). The full-window (−0.164) first stage is 0.0855 (0.020) and reduced form −0.0136 (0.005); the preferred-window (−0.190) first stage is 0.0851 (0.020) and reduced form −0.0154 (0.006).
The panel is unbalanced for two reasons: the ACS provides no sub-state geography for 2001–2004, so enrollment is observed in 2000 and 2005–2015; and the QWI series for Arkansas, Kentucky, and Wyoming begin after 2000, dropping 37 commuting-zone cells in that year (the three states Cascio and Narayan exclude altogether). This yields $N = 3{,}695$ for 2000–2015 and $N = 3{,}384$ excluding 2005.

Table 13: Fracking design: robustness (preferred window)

Variant	2SLS θ	SE	First-stage F	N
Play-specific timing[a]	-0.220^{**}	(0.095)	22.0	3,384
Barnett/Fayetteville CZs dropped	-0.207^{**}	(0.092)	20.2	2,910
Fringe trim ($< 5\%$ overlap zeroed)	-0.189^{**}	(0.084)	21.6	3,110

Notes. 2SLS coefficient on log expected income. CZ and state-by-year fixed effects; CZ-clustered standard errors (311 clusters, except 291 in the Barnett/Fayetteville-dropped row).
[a] Late plays interacted with a post-2008 indicator; the early-booming Barnett and Fayetteville plays interacted with a post-2003 indicator as a control; boom-long window excluding 2005.

Housing-boom design: full results and benchmark

Table 15 reports the housing-boom design. Column (1) reproduces the published Charles et al. (2018) attendance-on-housing-shock coefficient exactly, which validates the merge and construction pipeline. Column (2) instruments the change in the young low-education wage with the housing-demand instrument, and column (3) instruments the change in expected income, the object that enters the cross-design comparison of Table 4. The expected-income outcome here is an attainment stock (the any-college share of 18-to-25-year-olds in 2007, cohorts only partially exposed to boom-era entry decisions), which mechanically dilutes a flow enrollment response toward zero, so this estimate is best read as a lower bound on the magnitude, and its proximity to the fracking flow estimates is reassuring rather than mechanical.

Table 14: Fracking design by sex (preferred window)

	Pooled	Men	Women
First stage: expected income	0.085***	0.121***	0.033**
	(0.020)	(0.022)	(0.015)
Reduced form: enrollment	−0.0154***	−0.0176**	−0.0107
	(0.0058)	(0.0081)	(0.0074)
2SLS: enrollment on expected income	−0.190**	−0.150**	−0.307
	(0.084)	(0.073)	(0.251)
First-stage F (2SLS sample)	21.8	37.5	7.6

Notes. CZ and state-by-year fixed effects; CZ-clustered standard errors (311 clusters); N = 3,110 (reduced form and 2SLS), N = 3,732 (first stage). The female first stage is weak, so the female 2SLS is not interpreted.

Table 15: Housing-boom design (MSA level, 2000–2007 long difference)

	(1) Benchmark attend on shock	(2) 2SLS Δ ln wage	(3) 2SLS Δ ln exp. income
Housing demand shock	−0.0200***		
	(0.0065)		
Δ ln wage (young, low-ed)		−0.230***	
		(0.075)	
Δ ln expected income			−0.129***
			(0.044)
First-stage F		14.2	14.9
Observations (MSAs)	275	275	275
Clusters (states)	48	48	48

Notes. Charles et al. (2018) population weights and controls; standard errors clustered by state (48 clusters). Column (1) is the published Table 3 benchmark. Sensitivity of column (2): unadjusted wage −0.243*** (0.089); 18–34 age band −0.242*** (0.076) adjusted and −0.280*** (0.094) unadjusted; unweighted −0.167 (0.113), insignificant, consistent with the authors' own finding that their weighting matters. With 48 clusters and a state-level instrument component, a wild-cluster bootstrap is advisable if column (3) is ever quoted on its own.

China-shock design: full results and benchmark

Table 16 reports the China-shock design by sex, with its first stages, and Table 17 reproduces the published reduced form in its own units. The pooled estimate is −0.171; the design is the only one in the range with a strong female first stage, and it gives a female coefficient

nearly twice the (insignificant) male one, matching the published gender gradient.

Table 16: China-shock design by sex (2SLS and first stage)

	Pooled	Men	Women
First stage: exp. income on instrument	−0.0231	−0.0198	−0.0298
	(0.0043)	(0.0049)	(0.0062)
2SLS: enrollment on exp. income	−0.171***	−0.094	−0.206***
	(0.046)	(0.059)	(0.068)
First-stage F	28.9	16.4	22.8
Observations	1,444	1,444	1,444
Clusters (state)	48	48	48

Notes. Stacked panel of two ten-year-equivalent long differences (1990–2000 and 2000 to 2006–07), 722 commuting zones, Autor et al. (2013) controls, region dummies, and a window fixed effect; ADH population weights; standard errors clustered by state (t with 47 degrees of freedom). The endogenous variable is the change in log expected income of 18-to-29-year-olds with a high-school education or less; the instrument is the growth of Chinese exports to eight other high-income countries. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 17: China-shock design: published-benchmark reproduction

	All	Men	Women
Import exposure ($1,000/worker)	0.640***	0.358*	0.923***
	(0.221)	(0.194)	(0.316)
Published (FEDS Table 3)	0.88	0.67	1.10

Notes. 2SLS of the 18-to-25 any-college enrollment share (×100) on instrumented import exposure, the object reported by Conlisk et al. (2022). The reproduction matches the published sign, order of magnitude, significance pattern, and gender ordering (women roughly double men). The level runs about 16 to 47 percent below the published coefficients (16 percent for women, 27 percent pooled, and 47 percent for men) because those link individuals to their commuting zone five years earlier, a linkage not reconstructable from public microdata, whereas our residence-based assignment attenuates against enrollment-driven migration. This is a documented deviation, not a failed replication.

A second manuscript version of the same design (Ferrière, Navarro, and Reyes-Heroles, Table 3, column 2) reports a pooled coefficient of 0.878 (0.192), so the two published estimates of this design bracket our reproduction of 0.640.

Three further checks are worth recording. First, identification is entirely in the stacked panel: the single-window first stages are weak ($F = 0.9$ in 1990–2000, $F = 1.7$ in 2000–07), and the window-specific 2SLS coefficients (+0.23 and −0.55, both insignificant) are reported

only to document that pooling the two windows with a window fixed effect is what identifies θ. Second, a timing placebo that regresses the 1990–2000 changes on the future (2000–07) instrument, conditional on the contemporaneous instrument, finds a clean enrollment coefficient (-0.0004, $p = 0.59$) and a marginal opposite-signed pre-drift in expected income ($+0.0048$, $p = 0.09$) about a fifth of the first-stage magnitude; adjusting the first stage for that drift gives a trend-continuation bound of $\theta = -0.142$, inside the range. Measurement variants leave the estimate inside the bracket: restricting the income measure to the not-enrolled gives -0.207 (0.064), and an 18-to-25 outcome gives -0.175 (0.057). Third, because the second window coincides with the housing-boom window, we verify that the two designs are distinct: the import-exposure and housing-demand instruments are only weakly correlated across the commuting zones they share (-0.16 to -0.17), and on the matched sample the China estimate is essentially unchanged (-0.178, 0.059); conditioning on the housing shock does not pull it toward zero, though the conditional first stage becomes weak, so the two are not independent confirmations in the shared window even though the China design is clearly not the housing design in reverse.

Residency harmonization (born-in-state outcome)

The designs differ in residency concept: the housing design restricts to residents born in their state of residence (CHN's `_same` sample), whereas the fracking and China designs are residence-based over all residents. To examine whether this difference bears on the cross-design comparison, we harmonize on both sides. We recompute the fracking and China outcomes under the born-in-state restriction (`BPL` equal to `STATEFIP`), applied to the outcome only (the income measures stay market-level), with weights, fixed effects, instruments, and standard-error conventions unchanged; and we run the housing design without the born-in-state restriction using CHN's `_main` file. Each recomputed outcome uses the same enrollment-share formula, numerator and denominator, and the same extract and crosswalks as the main outcome.

Within each design the residency concept moves the estimate only modestly: the housing and fracking shifts are statistically indistinguishable from zero ($p = 0.56$ and 0.52) and the China shift is marginal ($p = 0.066$, significant at the ten-percent level only), so no shift changes any qualitative conclusion, although the China born-in-state value -0.107 lies just outside the bracket $[-0.13, -0.23]$. We do not read the differenced-outcome gaps as estimates of a migration contribution, since the born-in-state share can itself respond to the shock, and a born-in-state restriction is not the published designs' five-year location linkage: students who leave an exposed commuting zone to enroll elsewhere are lost from its numerator

Table 18: Residency harmonization: born-in-state outcome

Design	All residents	Born in state	Difference
Housing boom (stock)	-0.100^{*} (0.051)	$\mathbf{-0.129}^{***}$ (0.044)	-0.028 (0.048)
Fracking (flow)	$\mathbf{-0.190}^{**}$ (0.084)	-0.164^{*} (0.098)	-0.026 (0.040)
China shock (flow)	$\mathbf{-0.171}^{***}$ (0.046)	-0.107^{*} (0.057)	-0.065 (0.034)

Notes. 2SLS θ under an all-residents outcome and under the born-in-state restriction (`BPL=STATEFIP`), applied to the outcome only. **Bold** marks each design's headline estimate, the one reported in Table 4: all residents for fracking and China, born in state for housing, since Charles et al. (2018) publish the `_same` sample. The all-residents housing cell uses CHN's `_main` file, whose precise group-quarters universe cannot be verified from the package. First-stage F statistics are unchanged within design (14.9 housing, 21.8 fracking, 28.9 China), since the restriction touches only the outcome; the median commuting-zone born-in-state share of the 18-to-24 weight is 0.67. The difference column reports each design's *headline estimate minus its harmonization alternative*, applying the same first stage to the differenced outcome, with its own standard error ($p = 0.56$ housing, 0.52 fracking, 0.066 China). Because the housing headline is the born-in-state cell while the other two headlines are the all-residents cells, the housing difference is taken in the opposite left-right direction from the other two rows; all three are negative under the common convention.

The cross-language Stata pass for these rows is pending.

under either outcome, so the born-in-state estimate inherits the same enrollment-migration attenuation as the main one. The harmonization bears mainly on the flow-versus-stock reading: on a consistent unrestricted basis the stock estimate (-0.100) still lies below both flow estimates (-0.190 and -0.171), so the dilution reading survives the harmonization, which removes a potential confound of that reading rather than adding evidence for it. Cross-design differences remain untested, since the designs differ in geography, period, and weighting, and the flow-versus-stock caveat continues to apply. Residence-based sampling remains the main specification for fracking and China, with born-in-state as the appendix harmonization, because the calibration target (-0.190) sits on the audited residence-based estimates. CHN's `_main` attendance-on-shock benchmark is flatter and noisier than their published `_same` one (-0.0156 (0.0090) against -0.0200 (0.0065)), so nothing in the package contradicts their sample choice.

D Target Construction and Measurement

This appendix gives the construction of every calibration target of Section 5. All targets use the pooled 2010–2022 window and are deflated to 2018 PCE dollars, with dollar quantities in model units of $\mu = \$10{,}000$.

Externally fixed parameters

Cost curvature. We regress log(cost of attendance) on log(SAT) across pooled four-year public and nonprofit institutions in the College Scorecard, weighting by undergraduate enrollment, in nominal terms with no year fixed effects. The pooled slope is $\widehat{\zeta} = 1.82$ (SE 0.02, $N = 17{,}302$, $R^2 = 0.26$). For the pecuniary quality optimum to be interior, cost must be convex enough in quality relative to the human-capital return, $\zeta > \gamma(\nu+1)/\nu$; the estimated value satisfies this comfortably. Otherwise the marginal return to quality would dominate its marginal cost at every q and the optimum would run to the grid ceiling.

Pell maximum. The federal Pell Grant is the largest need-based grant program in the United States, and its maximum annual award is set by appropriation. Averaging the real 2018-dollar maximum over the 2010–2027 award years and multiplying by four for a bachelor's degree gives about \$24,000, or $\Phi = 2.4\,\mu$ (U.S. Department of Education, 2018; Congressional Research Service, 2018).

Federal loan limit. The \$31,000 aggregate cap, of which up to \$23,000 may be subsidized, is the Direct/Stafford limit for dependent undergraduates in the Federal Student Aid Handbook, fixed in nominal terms since 2008 and hence about \$31–32k in 2018 dollars over the target window. Model households finance college out of parental wealth and so face the dependent rather than the independent limit. Lochner and Monge-Naranjo (2011) impose a comparable statutory cap, which they set at \$35,000, a value chosen to lie between the dependent-undergraduate and independent/graduate limits.

Income-driven repayment and the CES elasticity. The repayment share $\varsigma = 0.10$ is the PAYE/REPAYE marginal rate and the income floor $\bar{y} = 36.42\,\mu$ is one and a half times the single-person poverty guideline aggregated to the period (Ji, 2021); both are statutory rather than estimated. The CES elasticity $\eta = 0.333$ follows Jones (2014), whose baseline elasticity of substitution between more- and less-educated labor is about 1.5; under our normalization the elasticity of substitution is $\sigma_E = 1/(1-\eta)$, so $\eta = 0.333$ delivers that value.

Targeted moments

Student debt. NPSAS variable `BORAMT1`, cumulative undergraduate borrowing, taken over all undergraduates enrolled in the wave. Nominal wave means are deflated to 2018 dollars by the calendar-year PCE price index and the 2012 and 2016 waves are equal-weight pooled.

Mean Pell award. Constructed from IPEDS Student Financial Aid (2010–2021, four-

year public and nonprofit institutions) as the product of the Pell receipt share (34.2%), the average annual award ($4,504), and four years of enrollment, giving about $0.617\,\mu$.

Labor-market moments. The CPS moments use the IPUMS ASEC extract, survey years 2010–2022. The college track is a BA or higher and the high-school track is a high-school diploma (rather than merely the absence of a BA). Full-time full-year requires at least 50 weeks of full-time work and a nominal weekly wage above $50, a measurement-error trim applied symmetrically to the model moment as a period earnings floor; cells span ages 25–64 (young 25–44, old 45–64), and lifecycle growth and the variance pool both sexes.

Wealth. SCF summary-extract files, waves 2010–2022; wealth is financial wealth net of student loans; singles are single-headed households; per-adult retirement assets divide married households' value by two.

Attendance by wealth quintile. Parental net worth is the NLSY97 1997 household measure; quintiles use the 1997 cross-sectional weights on the pooled sample, and each gender's profile is rescaled so its equal-weight quintile mean equals the CPS attainment target, matching the equal-mass bins the model implies.

Quality-earnings slope. The slope of 1.212 (SE 0.020) is a weighted least-squares regression of log median earnings ten years after entry (College Scorecard) on log average SAT, weighted by undergraduate enrollment, with year fixed effects, over four-year public and private nonprofit institutions ($N = 8{,}920$ school-years). Two choices are consequential and are recorded here because the target is sensitive to them: the same specification yields 1.18 with median six-year earnings and 1.45 with mean ten-year earnings. The reported target uses the median rather than the mean and the ten-year rather than the six-year horizon.

SAT quality gap. The female-to-male SAT ratio weights each institution by female and by male undergraduate enrollment separately, giving 0.988 pooled over 2010–2022. As external validation, the same gap built from student-level California FAFSA records (roughly 1.6 million filers, each linked to the institution attended and its average SAT) gives women 1,151 against men's 1,168, a ratio of 0.985, within three-tenths of a percentage point of the Scorecard figure despite differing in coverage, unit of observation, and weighting.

Initial distribution. From the SAT table of the NCES *Digest of Education Statistics* (National Center for Education Statistics, 2024), z is the mean total SAT divided by 1000, and $\mathrm{Var}(\log z)$ is recovered from the coefficient of variation by the delta method, $(\mathrm{SD}/\mathrm{mean})^2$. Ability in the correlation target is the age-adjusted CAT-ASVAB math-verbal percentile in the NLSY97, and the target is its weighted rank (normal-scores) correlation with 1997 parental net worth, over all respondents with both measures ($n = 5{,}413$). The pooled estimate is 0.365, essentially equal across genders (0.366 each).

The matriculation semielasticity conversion. Section 5 states the intuition; the formal statement is as follows. The data object is the change in the enrollment share per log point of expected income, the earnings of high-school-or-less workers times their employment rate. The model instead perturbs the wage w^H and measures the enrollment response per log point of the wage. Under GHH there is no income effect and hours are $n = \big((1-\tau)\, w\, h/\psi\big)^{1/\nu}$, so earnings satisfy $\mathrm{d}\ln y/\mathrm{d}\ln w = 1 + 1/\nu = 1.5$: half the response runs through the Frisch hours margin, which plays the role the employment rate plays in the data. Dividing the model's per-log-wage response by 1.5 therefore puts it in the same units as the estimate, and the constant elasticity makes the conversion exact.

E Model solution algorithm

This appendix records how the model is solved, in enough detail to reimplement. The model is solved by a single backward pass per lifecycle stage, then forward simulation, then moment construction; continuations are smooth, so backward induction is exact and no inner fixed point is required within a stage. Two conventions hold throughout. *In this appendix only, primes denote the next lifecycle stage*, so V', V'_a, V'_h are the continuation value and its envelopes on the next stage's grid. The GHH composite good is $\tilde{c} \equiv c - \psi^{g,e}(1-s)\, n^{1+\nu}/(1+\nu)$, with marginal utility $u_c = \tilde{c}^{-\rho}$, and hours $n(h)$ are always the closed form (22), which is why the endogenous-gridpoints (EGM) inversions below are analytic.[9]

Outer structure

The solution is assembled gender by gender. For each $g \in \{f, m\}$:

1. Build the grids described below.
2. For each track $e \in \{C, H\}$, solve the old-period value $V^{o,g,e}(h, a)$ by a single EGM pass, then the young-period value $V^{y,g,e}(h, a)$ by an EGM pass wrapped in the s-FOC root-find.
3. Solve the pre-period quality choice $q^{*,g}(z, a_0)$ and the conditional college value $V^{C,g}(z, a_0)$.
4. Compute the high-school value $V^{H,g}(z, a_0) = V^{y,g,H}(z, a_0)$ by interpolation.
5. Pin the level of the taste schedules $\kappa^g(a_0)$ by the attainment inner loop, and form the matriculation probabilities $P_C^g(z, a_0)$ from the logit (39).
6. Forward-simulate the stationary distribution across stages; compute the aggregates (matriculation, mean signed debt, mean earnings, hours, retirement assets, and the effective labor supplies H_C, H_H); and recover (A_C, A_H) from the factor-demand conditions (40).

[9] The preference specification also carries a Stone-Geary subsistence term $\bar{c}$, set to zero, so within-period preferences are pure GHH. The income-driven-repayment branch of (34) enters only the young-college backward pass, as an upper envelope over a one-dimensional discharge value, and leaves the within-period GHH block and the EGM inversions below unchanged.

Grids

Assets. Two grids span the asset dimension. The nonnegative grid $\mathbf{a}_+ = [0, a_{\max}]$ ($n_a = 100$ log-spaced points, $a_{\max} = 2000\,\mu$, first positive node $0.02\,\mu$) serves the young-high-school state, the old and retirement problems, the EGM continuation axis (where $a' \geq 0$), and the parental-wealth axis a_0. The young-college state grid adds the debt region as 50 uniform nodes on $[-d_{\max}, 0)$, with the statutory cap $-d_{\max} = -3.1\,\mu$ exactly on-grid, so the whole feasible range of entry indebtedness is resolved at fixed spacing. A uniform debt grid, rather than a log-spaced one running from an uncapped floor (which would leave only about two nodes on the active debt region), keeps the debt moments free of grid-interpolation error; they converge by ten nodes (Table 19).

Effort. Log-spaced on $[10^{-5}, \bar{s}]$ ($n_s = 75$ points), with ceiling

$$\bar{s} = \min\{1,\ 1.10\, s_\infty(h_{\min})\},$$

where $s_\infty(h)$ is the unconstrained optimum of (28). Under the single-crossing bound $\alpha < \nu/(1+\nu)$, $s_\infty(h)$ is non-increasing in h, so a single root-find at $h_{\min}$ gives the ceiling; the floor 10^{-5} avoids the $(hs)^{\alpha-1}$ singularity in the s-FOC.

Quality. Linear on $[0.6, 2.4]$ ($n_q = 75$ points); the bounds are the SAT composite range on the $q = \text{SAT}/1000$ scale, and the cap $q_{\max} = 2.4$ overrides the slacker theory ceiling $1.15\, q^*_\infty(z_{\max}, m_C)$. **Ability.** The nominal range $[0.8, 2.4]$ is the SAT-scale anchor, but the grid extends the floor down to the four-sigma point of the calibrated lognormal, $\exp(\mu_z - 4\sigma_z) \approx 0.39$, on $n_z = 45$ linear nodes, so the discretized initial distribution carries the full lognormal support (clipped mass about 3×10^{-5}) and the simulated ability moments equal the analytic moments in Table 6. Clipping instead at the literal SAT floor 0.8 would truncate about 27% of the calibrated ability mass onto the boundary cell.

Graduation shock. $\log \varepsilon \sim \mathcal{N}(-\sigma^2/2, \sigma^2)$ is integrated by Gauss-Hermite quadrature with $n_\varepsilon = 7$ nodes $\{\varepsilon_j, w_j\}$, the nodes shifted so that $\sum_j w_j \varepsilon_j = 1$.

Human capital. Three log-spaced grids ($n_h = 150$ points each): the young-college grid spans the convex hull of $\{z\,\varepsilon_j(1+q^\gamma)\}$; the young-high-school grid equals the z-grid, since $h = z$; and the old grid is the image of the young grid under the law of motion (24), sized separately for each track (h^C_{old} and h^H_{old}) from its own scale B^e so high-school resolution is not spent spanning the college ceiling. Each grid uses $n_h = 150$ points; convergence in n_h is reported in Table 19.

Table 19: Grid convergence of the gender debt gap

Active debt-region nodes	2	10	25	50
Gender debt gap (μ)	0.405	0.4530	0.4531	0.4531

Notes. Gender debt gap at a fixed parameter vector, chosen to isolate grid effects rather than to match the data, as the number of uniform nodes on the active debt region $[-d_{\max}, 0)$ is varied. The moment converges to within 0.1% by ten nodes; debt levels converge to four decimals and discharge shares to about one percent. At the finer resolution $n_h = 225$, $n_q = 150$ the targeted moments move by at most about 3%. The calibration uses 50 debt nodes, $n_h = 150$, $n_q = 75$, $n_z = 45$, and $n_\varepsilon = 7$.

Old period

Single-pass EGM on $(h, \mathbf{a}_+)$ with the closed-form retirement value (25). The iterated objects are the value V^o and its envelopes (V^o_a, V^o_h).

1. **Hours.** $n(h)$ from (22); independent of wealth.

2. **Euler inversion $a' \to \tilde{c}$.** The Euler equation $u_c = \beta V^R_a(a')$ with (25) inverts in closed form for the composite good,

$$\tilde{c}_{\text{endog}}(h, a') = \left(\beta\, R\,(a' + \bar{a}_{VR})^{-\rho}\right)^{-1/\rho}, \qquad c_{\text{endog}} = \tilde{c}_{\text{endog}} + \psi^{g,e}\,\frac{n(h)^{1+\nu}}{1+\nu}. \tag{E1}$$

3. $a' \to a$**.** The budget constraint in (26) defines the endogenous current asset,

$$a_{\text{endog}}(h, a') = \frac{c_{\text{endog}} + a' - (1-\tau^{g,e})\, w^e\, h\, n(h)}{1+r}, \tag{E2}$$

   and inverting it gives the policy $a'(h, a)$ on the exogenous grid $\mathbf{a}_+$ by linear interpolation, clamped to $a' \geq 0$.

4. **Recover c and V.** From the budget constraint and the flow return,

$$c(h,a) = (1+r)\,a + (1-\tau^{g,e})\, w^e\, h\, n(h) - a'(h,a), \qquad V^{o,g,e}(h,a) = u^{g,e}(c,n) + \beta\, V^R\big(a'(h,a)\big). \tag{E3}$$

5. **Envelopes.** Record, for the young stage, the marginal values of assets and human capital carried into old age,

$$V^o_a = (1+r)\, u_c, \qquad V^o_h = (1-\tau^{g,e})\, w^e\, n\, u_c. \tag{E4}$$

Because $V_a^R \to \infty$ at $a' = -\bar{a}_{VR}$, optimal old-period assets are bounded above zero in every state, so the borrowing floor never binds and restricting the old grid to $\mathbf{a}_+$ is without loss.

Young period

EGM over a' on $\mathbf{a}_+$ conditional on s, wrapped in a root-find of the study first-order condition (28), on the cube (h, s, a). Steps 1–4 run at each s on the effort grid; step 5 selects s^*.

1. **Project the continuation $h \to h'$.** Form $h'(h, s) = B^e (hs)^\alpha + (1 - \delta) h$ from (24) and interpolate the old value and envelopes (V^o, V_a^o, V_h^o) onto h'.

2. **Hours.** $n(h)$ from (22); independent of s.

3. **Euler inversion given s.** With V_a^o evaluated at h', the intertemporal condition $u_c = \beta V_a^o(h', a')$ inverts in closed form for the composite good,

$$\tilde{c}_{\text{endog}} = \left(\beta\, V_a^o(h', a')\right)^{-1/\rho}, \qquad c_{\text{endog}} = \tilde{c}_{\text{endog}} + \psi^{g,e}(1 - s)\, \frac{n(h)^{1+\nu}}{1+\nu}, \tag{E5}$$

   and the budget constraint in (27) gives the endogenous asset,

$$a_{\text{endog}} = \frac{c_{\text{endog}} + a' - (1 - \tau^{g,e})\, w^e\, (1 - s)\, h\, n(h)}{1 + r}, \tag{E6}$$

   back-interpolated onto the young state grid $\mathbf{a}_y$ (which includes debt states) with the continuation a' clamped to $\mathbf{a}_+$.

4. **Feasibility.** With cash-on-hand $\text{coh}(h, s, a) = (1+r)a + (1-\tau^{g,e}) w^e (1-s) h\, n(h)$, flag the cells that admit no feasible consumption,

$$\text{feas}(h, a) = \mathbf{1}\Big[\max_s\ \text{coh}(h, s, a) > c_{\min}\Big], \qquad c_{\min} = 10^{-10}.$$

   Infeasible cells receive the sentinels $V = -10^{300}$, $V_a = 10^{-12}$, $V_h = 0$ and are excluded from the forward scatter, so no garbage policy can place mass at a grid edge.

5. **s-FOC.** On the effort grid, evaluate the residual of the study first-order condition (28),

$$R_s(h, s, a) = u_c \left[\psi^{g,e} \frac{n^{1+\nu}}{1+\nu} - (1 - \tau^{g,e})\, w^e\, h\, n\right] + \beta\, V_h^{o,g,e}(h', a')\, \alpha B^e (hs)^{\alpha-1} h, \tag{E7}$$

   with u_c evaluated at the s-conditional (c, n), and find the root $s^*(h, a)$ with a bracketed finder: an exact zero is returned directly, an all-same-sign column returns the endpoint

nearest zero, a single crossing is linearly interpolated, and multiple crossings are resolved by the highest V. Interpolate the policies (a', c, s^*) and the envelopes (V_a^y, V_h^y) at s^*.

The two tracks share this pass and differ only in the prices and technology they carry, $(w^e, \tau^{g,e}, \psi^{g,e}, B^e)$, and in the entry state, $(h, a) = (z, a_0)$ for high school.

Income-driven repayment

The young-college value is the upper envelope (34) of full repayment and the discharge branch. The full-repayment value is the young EGM pass above, solved on the debt-carrying grid $\mathbf{a}_y$. The discharge value $V_{\text{IDR}}^{y,g,C}(h)$ solves the same young problem with the repayment cap (33) in place of full amortization and the residual balance forgiven at the transition to old age; because the forgiven balance leaves the budget, this branch does not depend on the debt level and is solved once per h-node, one dimension cheaper than the full branch. For each debt state the solver takes the pointwise maximum of the two values and records which branch is active, and that indicator, aggregated in the forward pass, is the model's loan-discharge rate.

Pre-period: quality choice

The quality choice is solved on the (z, a_0) grid. For each (z, a_0) and each quality node q_k:

1. **Entry state.** Compute entry assets $a_{\text{entry}}(q_k) = a_0 - k_{\text{eff}}(q_k, a_0)$ from (32); mark q_k infeasible if it violates the loan cap, $-\min\{0, a_{\text{entry}}(q_k)\} > d_{\max}$ (equivalently $q_k > \bar{q}(a_0)$), which is also the lower edge $-d_{\max}$ of the debt grid. For each shock node ε_j, form $h^{(j)} = z\,\varepsilon_j(1 + q_k^{\gamma})$ and interpolate $(V^{y,g,C}, V_h^{y,g,C}, V_a^{y,g,C})$ at $(h^{(j)}, a_{\text{entry}}(q_k))$.

2. **Integrate over ε.** Aggregate to the conditional college value (36),

$$V^{C,g}(z, a_0, q_k) = \sum_j w_j\, V^{(j)}, \tag{E8}$$

   on the feasible range $q_k \le \bar{q}(a_0)$ set by the loan cap.

3. **Quality choice.** The chosen quality is the value-maximizing node on the feasible grid, $q^*(z, a_0) = \arg\max_{k:\, q_k \le \bar{q}(a_0)} V^{C,g}(z, a_0, q_k)$, resolved by direct comparison of the conditional values (no first-order condition is used; the objective is not single-peaked under income-driven repayment), and the value of attending is $V^{C,g}(z, a_0) = V^{C,g}(z, a_0, q^*(z, a_0))$.

A fused kernel performs the batch interpolation and the ε-reduction in a single pass, producing the conditional values with no large intermediates.

College attendance

For each (z, a_0), the high-school value $V^{H,g}(z, a_0) = V^{y,g,H}(z, a_0)$ is read off the young-high-school solution by interpolation; since the a_0 axis equals $\mathbf{a}_+$ by construction, no re-interpolation is needed on that axis. Matriculation is the numerically stable logit (39),

$$P_C^g(z, a_0) = \text{sigmoid}\left(\frac{V^{C,g}(z, a_0) + \kappa^g(a_0) - V^{H,g}(z, a_0)}{\sigma_\kappa}\right),$$

with the wealth-dependent shifter $\kappa^g(a_0)$ (a step function over parental-wealth quintiles) built by a shared helper.

Forward distribution

The stationary distribution is built by propagating $F(z, a_0)$ forward across the working stages, carrying the permanent ability z alongside (h, a). Off-grid images are distributed by linear-lottery weights.

1. **Discretize.** Represent $F(z, a_0)$ from (20) on $\mathbf{z} \times \mathbf{a}_+$ as $\mu_0(z, a_0)$ by a bivariate-lognormal cell-mass rule.

2. **Pre-period to young.** Split each cell's mass by the matriculation probability: mass $(1 - P_C^g)\,\mu_0$ goes to the high-school distribution at $(z,\ h = z,\ a = a_0)$, and mass $P_C^g\,\mu_0$ is placed at the chosen quality $q^*(z, a_0)$ and spread across the shock nodes, landing in the college distribution at

$$\big(z,\ h = z\,\varepsilon_j(1 + q^{*\gamma}),\ a = a_{\text{entry}}(q^*, a_0)\big) \quad \text{with weight } w_j.$$

3. **Young to old.** Apply the policies (s^*, a'^*): each cell (z, h, a) maps to $\big(z,\ h' = B^e(hs^*)^\alpha + (1 - \delta)h,\ a' = a'^*(h, a)\big)$ on the track-split old grid. Cells flagged infeasible in the backward pass ($V^y \le -10^{200}$) are excluded so their sentinel policies cannot contaminate the old distribution.

4. **Old to retirement.** Apply the old-period asset policy and collapse the human-capital dimension by a one-dimensional lottery, giving a distribution over retirement assets

alone; this is the object underlying the retirement-wealth target, $\mathbb{E}[a \mid \text{retire}, g, e]$, rather than average assets during the old working period.

5. **Aggregate.** Moments are mass-weighted sums over the young, old, and retirement distributions, with debt measured as $-\min\{0, a\}$ on the young-college entry distribution.

Convergence and cost

The policy solution has no inner fixed point; continuations are smooth, so backward induction is exact. Robustness rests on a consumption floor $c_{\min} = 10^{-10}$ for feasibility flagging, the infeasibility sentinels above, and the bracketed root rules for the s-FOC and the pre-period derivative. The dominant cost is the young backward step, $O(n_h\, n_s\, n_a)$; the pre-period adds a factor $n_z\, n_\varepsilon$ over its base cost, and the income-driven-repayment envelope adds a one-dimensional discharge solve per h-node. The backward passes parallelize over the h-axis.